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Algorithmic Methods of Data Mining

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1 Introduction to data mining

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1.1 Course organization

The course has two parts. The first part brings everyone to the same level, especially those who have not studied computer science before: basic computer architecture, data structures (different problems call for different ways of storing data), algorithms (how to solve a problem quickly, or even optimally, and how to measure its complexity), and programming, practised in the lab, in the homeworks and in the hackathon. The second part covers data mining topics: text mining, clustering and graphs.

1.2 What is data mining?

Huge amounts of data are produced in every field: social networks, for instance, both contain and generate an enormous quantity of data. These data hide patterns that we would like to find, but the amount is far too large to inspect by hand. Data mining tools let us explore data that we could not otherwise process.

Definition 1.1: Data mining

Data mining is the use of efficient techniques for the analysis of very large collections of data and the extraction of useful and possibly unexpected patterns in data.

We will meet two kinds of tasks:

- **Descriptive tasks** summarize the data and reveal their structure, for example groups of similar objects.

- **Predictive tasks** use the data to predict something unknown, for example whether a person will click on an advertisement.

1.3 Structured, unstructured and semi-structured data

Data come with different amounts of organization.

Definition 1.2: Structured, unstructured and semi-structured data

- *Structured data* have a clean, fixed form. For example, a university's student records store a name, a surname and a student ID for every student. These are the best data we can hope for.
- *Unstructured data* have no predefined form, for example text and pages found on the internet. Most of the data we meet are of this kind.
- *Semi-structured data* lie in between: they have some structure, but it can be modified freely. Spreadsheets (Excel) and XML files are typical examples.

Note: Data cleaning

Data must be cleaned before they are analysed: if the data are dirty, the algorithm will not work. Cleaning used to be very hard to do by hand; today software tools help, but we should still check that the result is correct. Semi-structured data are especially prone to errors, because their structure is easy to break.

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1.4 Types of data

Data are very heterogeneous. Common types include:

- **Table (record) data**, such as spreadsheets; also images and time series.
- **Transaction data**: each record is a transaction between two parties, such as a credit-card payment or a shopping basket.
- **Document data**: Wikipedia pages, tweets, ...
- **Network data**: the web, where pages are linked to one another. Google works with these graphs, and Wikipedia's internal links form one as well.
- **Genomic sequences**: databases of DNA sequences, often used to find correlations with diseases.
- **Climate data**.
- **Behavioural data**: everything recorded about us, for example by smartphones.

1.5 Data as mathematical objects

1.5.1 Records, attributes and sparsity

In table data, each *row* describes an object (for example a person) and each *column* describes an *attribute* of the object.

Definition 1.3: Sparsity

A data table is *sparse* when most of its cells are empty (or zero). The *sparsity* of a table describes how many of its cells are filled, that is, whether many or few values are missing.

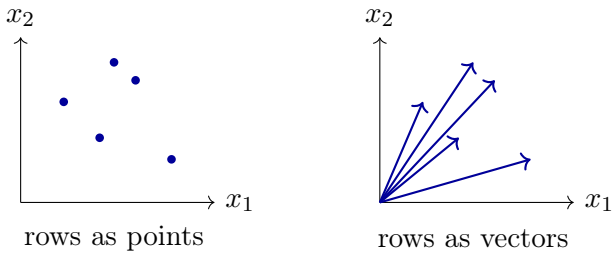
Attributes can be of different types:

- **Binary**: two possible values, such as yes/no.
- **Categorical**: values in a finite set of classes, such as a colour or a school grade like "good" or "bad".
- **Numeric**: numbers, either *discrete* (counts) or *continuous* (time, length, temperature).

We can also move between types: a numeric attribute can be turned into a categorical one by grouping values into classes, and categories can be encoded as numbers.

1.5.2 Numeric records as points and vectors

To analyse data with algorithms, we represent them as mathematical objects. Suppose a table has 2000 rows (people) and 7 numeric attributes. Each row is then an element of $\mathbb{R}^7$, which we can view either as a *point* in a 7-dimensional space or as a *vector* starting from the origin.



1.5.3 Document data

A text contains many words. We represent a document as a vector with one dimension per word of the vocabulary; each entry counts how many times that word appears in the document.

Example 1: Documents as word-count vectors

	word 1	word 2	word 3	⋯
Document 1	10	7	1	⋯
Document 2	15	5	0	⋯

The vocabulary is large, and each document uses only a small part of it. Most entries are therefore 0, so document data have a very high sparsity.

1.5.4 Transaction data

Each transaction has an identifier (TID) and the set of items it involves.

Example 2: A transaction table

TID	Items
1	bread, coke, milk
2	milk, apples
3	carrots
4	chocolate, honey

1.5.5 Ordered data and graph data

In *ordered data* the order of the entries is essential. A genomic sequence such as `GGTTCCGCTTCA`... is an ordered string of characters; time series are another example. In *graph data*, objects are nodes and relations are edges. The classic example is the web graph, whose nodes are pages and whose edges are links.

2

Analysis of algorithms

2.1 Problems and algorithms

A computational *problem* specifies which inputs are allowed and which output each input should produce. An *algorithm* is a finite sequence of steps that produces the required output for every input. Our first problem is finding the maximum of an array.

Problem: Max-Find

Input: an array $A = (A[1], A[2], \ldots, A[n])$ of n numbers.
Output: the maximum value in A.

Note: Indexing

In pseudocode we index arrays from 1, so $A[i]$ is the i-th element. In Python, indexing starts from 0: for $\vec{x} = (x_1, x_2, x_3)$ we have `x[0]` $= x_1$. The length of the array is `n = len(A)`.

Algorithm 2.1: Max-Find

```
1: function MaxFind(A)
2:     max ← A[1]
3:     for i = 1 to n do
4:         if max < A[i] then
5:             max ← A[i]
6:         end if
7:     end for
8:     return max
9: end function
```

The loop may also start from $i = 2$, since comparing $A[1]$ with itself is useless. For every algorithm we always ask two questions:

1. **Correctness:** does it solve my problem?
2. **Efficiency:** how fast is it?

2.2 Correctness

Testing an algorithm on some inputs is not enough: it may work only on the inputs we tried. To show that an algorithm is *wrong*, a single counterexample suffices. To show that it is *correct*, we need a proof, typically by induction on the number of steps performed.

Proposition 2.1: Correctness of Max-Find

After the iteration with index i, the variable max in Algorithm 2.1 equals

$$\max\{A[1], \ldots, A[i]\}.$$

In particular, when the loop ends ($i = n$), the algorithm returns the maximum of A.

Proof. We use induction on i.

Base case ($i = 1$): $max = A[1]$, which is the maximum of $\{A[1]\}$.

Inductive step: assume that after iteration i we have $max = \max\{A[1], \ldots, A[i]\}$. In iteration $i + 1$,

$$A = \big[\underbrace{A[1], A[2], \ldots, A[i]}_{\text{maximum}=max}, A[i+1], \ldots, A[n]\big].$$

If $A[i+1] > max$, then $A[i+1]$ becomes the new maximum; otherwise we keep the previous one. In both cases $max = \max\{A[1], \ldots, A[i+1]\}$. □

2.3 Efficiency: counting steps

How long an algorithm takes depends on many things: the program, but also the input, the programming language, the software, the operating system and the hardware. To compare algorithms, we ignore all of these and look only at the algorithm itself: we count how many elementary steps it performs.

Definition 2.1: Running time

The *running time* $T(n)$ of an algorithm is the number of steps it requires on an input of size n. Unless stated otherwise, we consider the *worst case*: the largest number of steps over all inputs of size n.

Example 3: Counting the steps of Max-Find

```
max ← A[1]                                   ▷ 1 step
for i = 1 to n do                            ▷ n steps
    if max < A[i] then                       ▷ n steps
        max ← A[i]                           ▷ k ≤ n steps
    end if
end for
return max                                   ▷ 1 step
```

Hence $T(n) = 2n + k + 2$. The number k of updates depends on the input:

- *Best case*: the maximum is $A[1]$, so the update never runs ($k = 0$).
- *Worst case*: the array is sorted in increasing order, so the update runs every time ($k = n$).

In the worst case,

$$T(n) = 2n + n + 2 = 3n + 2.$$

The same algorithm can be written with a `while` loop:

Example 4: Max-Find with a while loop

```
max ← A[1]                                   ▷ 1 step
i ← 1
while i ≤ n do                               ▷ n steps
    if A[i] > max then                       ▷ n steps
        max ← A[i]                           ▷ ≤ n steps
    end if
    i ← i + 1                                ▷ n steps
end while
return max                                   ▷ 1 step
```

In the worst case $T(n) = 4n + 2$.

Both versions are correct and do the same thing, and both running times are linear functions of n. The exact constants depend on how we count: initializing i is one more step, and the loop test runs once more when it finally fails. Such details change only the constants, which is why we will learn to ignore them.

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2.4 Asymptotic notation

In the previous section we counted the steps of two versions of Max-Find, one with a `for` loop and one with a `while` loop, and obtained $T(n) = 3n + 2$ and $T(n) = 4n + 2$. Which

one is faster? The exact numbers depend on how we count, so comparing them directly tells us little. What matters is how the running time changes as the input grows: if n doubles, both running times roughly double. In this sense, both *grow like* n.

Asymptotic ("big O") notation makes this idea precise. It ignores constant factors and lower-order terms: the $+2$ can be dropped, and $4n$ grows faster than $3n$ only by a constant factor, which makes no real difference.

Definition 2.2: O, Ω and Θ

Let $T, g : \mathbb{N} \to \mathbb{R}_{\geq 0}$.

- $T(n) = O(g(n))$ if there are constants $c > 0$ and n_0 such that $T(n) \leq c\, g(n)$ for all $n \geq n_0$: g is an *upper bound* on the growth of T.
- $T(n) = \Omega(g(n))$ if there are constants $c > 0$ and n_0 such that $T(n) \geq c\, g(n)$ for all $n \geq n_0$: g is a *lower bound*.
- $T(n) = \Theta(g(n))$ if both $T(n) = O(g(n))$ and $T(n) = \Omega(g(n))$: T grows exactly like g, up to constants.

Informally, O means "at most", Ω means "at least" and Θ means "exactly", always up to constant factors and for large n. The running time of Max-Find is $T(n) = \Theta(n)$: it grows like a linear function of n.

Proposition 2.2: Combining upper and lower bounds

$$T(n) = O(g(n)) \text{ and } T(n) = \Omega(g(n)) \iff T(n) = \Theta(g(n)).$$

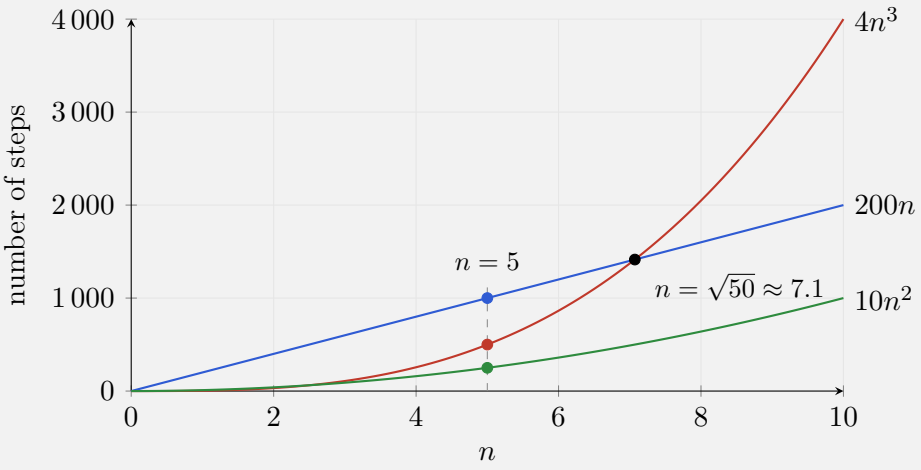
This is the asymptotic version of "$x \leq 3$ and $x \geq 3$ imply $x = 3$". For instance, $T(n) = O(n^3)$ and $T(n) = \Omega(n^3)$ give $T(n) = \Theta(n^3)$.

Upper bounds are not unique: if $T(n) = \Theta(n^2)$, then also $T(n) = O(n^2)$, $T(n) = O(n^3)$, $T(n) = O(n^3 \log n)$, and so on.

Example 5: The dominant term

Let $T(n) = 4n^3 + 10n^2 + 200n + 7$. How fast does this function grow?

The plot compares the three terms that depend on n.



For small n, a lower-order term can be the largest. At $n = 5$ we have

$$200n = 1000, \qquad 4n^3 = 500, \qquad 10n^2 = 250.$$

As n grows, however, the cubic term overtakes $200n$ at $n = \sqrt{50} \approx 7.1$ and dominates from then on, so it determines the running time.

Upper bound. For $n \geq 1$, every power of n in $T(n)$ is at most n^3, so

$$T(n) \leq 4n^3 + 10n^3 + 200n^3 + 7n^3 = 221\, n^3.$$

Hence $T(n) = O(n^3)$, with $c = 221$ and $n_0 = 1$.

Lower bound. All the terms are non-negative, so

$$T(n) \geq 4n^3.$$

Hence $T(n) = \Omega(n^3)$, with $c = 4$ and $n_0 = 1$.

Conclusion. Therefore, by Proposition 2.2, $T(n) = \Theta(n^3)$.

Other upper bounds. For $n \geq 1$,

$$T(n) \leq 221\, n^3 \leq 221\, n^4 \leq 221\, n^5,$$

so we also have $T(n) = O(n^4)$ and $T(n) = O(n^5)$. However, $T(n) \neq \Theta(n^4)$: since $T(n)/n^4 \leq 221/n \to 0$, no constant $c > 0$ gives $T(n) \geq c\, n^4$ for all large n.

When analysing an algorithm we are mostly interested in the *upper bound*, that is, in the worst case. Often we do not know exactly how many times a step runs. In Algorithm 2.1, we do not know in advance how many times the update $max \leftarrow A[i]$ will run, but in the worst case (a sorted array) it runs n times, so it never runs more than n times. Bounding each step from above gives an upper bound on $T(n)$.

2.4.1 Comparing growth rates

Proposition 2.3: Growth rates

For large n (that is, up to constants and eventually):

- $n^c \leq n^d$ for $c \leq d$;
- $\log^c n = (\log n)^c \leq n^d$ for all constants $c, d > 0$;
- $n^c \leq d^n$ for every constant c and every $d > 1$;
- more generally, $n^c \leq d^{an^b}$ for $d > 1$ and constants $a, b > 0$.

Consequently,

$$\begin{aligned} 1 \leq \log\log n \leq \log n \leq \log^2 n \leq \sqrt{n} \leq n \\ \leq n \log\log n \leq n \log n \leq n\sqrt{n} \leq n^2 \leq n^3 \leq 2^n. \end{aligned}$$

For example, $(\log n)^{10}$ and $\sqrt{n}$: the logarithm raised to any power always loses eventually, so $\sqrt{n}$ grows faster.

Note: The base of the logarithm does not matter

Changing the base of a logarithm only multiplies it by a constant:

$$\log_2 n = \frac{\log_{10} n}{\log_{10} 2} = \underbrace{\frac{1}{\log_{10} 2}}_{\text{constant}} \cdot \log_{10} n.$$

Asymptotically all bases are equivalent, so we simply write $O(\log n)$.

An exponential running time means that the algorithm is extremely slow: an algorithm with exponential running time is hopeless except for tiny inputs. Ideally we would like logarithmic or linear running times.

n	$\log_2 n$	$n \log_2 n$	n^2	2^n
10	3.3	33	10^2	$\approx 1.0 \cdot 10^3$
100	6.6	664	10^4	$\approx 1.3 \cdot 10^{30}$
1000	10.0	9966	10^6	$\approx 1.1 \cdot 10^{301}$

2.5 Lower bounds and optimal algorithms

Max-Find is correct and runs in time $\Theta(n)$. Could we do better, say $O(\sqrt{n})$? No: any correct algorithm must read every element, because an element it never reads could be the maximum. With n elements to read, no algorithm can beat $\Omega(n)$.

Definition 2.3: Lower bound and optimal algorithm

A problem has *lower bound* $\Omega(g(n))$ if every algorithm that solves it needs $\Omega(g(n))$ steps in the worst case. An algorithm is (asymptotically) *optimal* if its running time matches a lower bound for the problem.

Example 6: Max-Find is optimal

Max-Find (Algorithm 2.1) runs in $O(n)$, as we computed in Example 3, and the problem requires $\Omega(n)$, so Max-Find is *optimal*.

2.6 Prefix sums

Problem: Prefix sums

Input: an array A of length n.
Output: an array S of length n such that, for every i,

$$S[i] = \sum_{j=1}^{i} A[j].$$

For example, $A = [3, 6, -2, -5, 8]$ gives $S = [3, 9, 7, 2, 10]$.

2.6.1 A quadratic algorithm

The direct approach computes each sum from scratch.

Algorithm 2.2: Prefix sums, direct version

```
1: n ← length(A)                                  ▷ 1 step
2: for i = 1 to n do                              ▷ n steps
3:     S[i] ← 0                                   ▷ n steps
4:     for j = 1 to i do                          ▷ i times for the i-th element
5:         S[i] ← S[i] + A[j]                     ▷ i times
6:     end for
7: end for
8: return S                                       ▷ 1 step
```

The inner loop runs once for the first element, twice for the second, and n times for the last. Each of its two lines therefore runs

$$1 + 2 + 3 + \cdots + n = \sum_{i=1}^{n} i = \frac{n(n+1)}{2}$$

times, and

$$T(n) = 1 + n + n + 2 \cdot \frac{n(n+1)}{2} + 1 = n^2 + 3n + 2 = \Theta(n^2).$$

Note: A quicker upper bound

We do not need the exact sum. For every i, the inner loop runs at most n times, so

$$T(n) \leq 2n + n \cdot (2n) + 2 = O(n^2).$$

Bounding each loop by its largest possible number of iterations is often enough to obtain an upper bound.

2.6.2 Closing the gap

Is this algorithm optimal? No: the lower bound for the problem is $\Omega(n)$ (we must read every element), while our algorithm takes $\Theta(n^2)$. There is a *gap*, and we can try to close it by finding a better algorithm. The key observation is that each prefix sum extends the previous one:

$$S[i] = \sum_{j=1}^{i} A[j] = \sum_{j=1}^{i-1} A[j] + A[i] = S[i-1] + A[i].$$

Algorithm 2.3: Prefix sums, fast version

```
1: n ← length(A)
2: S[1] ← A[1]
3: for i = 2 to n do
4:     S[i] ← S[i − 1] + A[i]                    ▷ n − 1 times
5: end for
6: return S
```

Now $T(n) = \Theta(n)$, which matches the lower bound: the fast version is optimal.

2.7 An open gap: matrix multiplication

Let $A, B \in \mathbb{R}^{n \times n}$ and $C = AB$. Each entry is

$$C[i,j] = \sum_{k=1}^{n} A[i,k]\, B[k,j].$$

Computing one entry takes n multiplications, and there are n^2 entries, so the direct algorithm runs in $\Theta(n^3)$.

A lower bound is $\Omega(n^2)$: the output matrix C has n^2 values, and we must compute each of them. For many years, researchers have tried to close the gap between n^2 and n^3. The gap has been reduced, but it has not yet been closed.

Note: For reference

Strassen's algorithm (1969) multiplies two $n \times n$ matrices in $O(n^{\log_2 7}) \approx O(n^{2.81})$ operations. The best known exponent today is about 2.37, still above the lower bound 2.

3 Recursion

3.1 Fibonacci numbers

Definition 3.1: Fibonacci sequence

$$\text{fib}(n) = \begin{cases} 1, & n = 0 \text{ or } n = 1, \\ \text{fib}(n-1) + \text{fib}(n-2), & n \geq 2. \end{cases}$$

n	0	1	2	3	4	5	6	7	8	9
$\text{fib}(n)$	1	1	2	3	5	8	13	21	34	55

The definition is *recursive*: it defines $\text{fib}(n)$ in terms of smaller values. It translates directly into a recursive algorithm, one that calls itself.

Algorithm 3.1: Recursive Fibonacci

```
1: function FIBONACCI(n)
2:     if n = 0 or n = 1 then                          ▷ 1 step
3:         return 1                                    ▷ at most 1 step
4:     end if
5:     f1 ← FIBONACCI(n − 1)                           ▷ T(n − 1) steps
6:     f2 ← FIBONACCI(n − 2)                           ▷ T(n − 2) steps
7:     return f1 + f2                                  ▷ 1 step
8: end function
```

3.1.1 Running time of the recursive algorithm

Let $T(n)$ be the number of steps for input n. Counting the steps of Algorithm 3.1 gives a *recurrence*:

$$T(n) = \begin{cases} 2, & n = 0 \text{ or } n = 1, \\ T(n-1) + T(n-2) + 3, & n \geq 2. \end{cases}$$

The formula is recursive. How can we solve it? A first idea is to expand it. The recurrence holds for *every* $n \geq 2$, so we can apply it to the terms on the right as well: replacing n with $n-1$ gives $T(n-1) = T(n-2) + T(n-3) + 3$, and replacing n with $n-2$ gives $T(n-2) = T(n-3) + T(n-4) + 3$. In the computations below, each underlined term is replaced, in the next line, by the expression underlined in the same colour.

First expansion.

$$\begin{aligned} T(n) &= \underline{T(n-1)} + \underline{T(n-2)} + 3 \\ &= \underline{\big[T(n-2) + T(n-3) + 3\big]} + \underline{\big[T(n-3) + T(n-4) + 3\big]} + 3 \\ &= T(n-2) + 2T(n-3) + T(n-4) + 9. \end{aligned}$$

Second expansion. We expand each of the three terms again:

$$\begin{aligned} T(n) &= \underline{T(n-2)} + \underline{2T(n-3)} + \underline{T(n-4)} + 9 \\ &= \underline{\big[T(n-3) + T(n-4) + 3\big]} + \underline{2\big[T(n-4) + T(n-5) + 3\big]} \\ &\quad + \underline{\big[T(n-5) + T(n-6) + 3\big]} + 9 \\ &= T(n-3) + 3T(n-4) + 3T(n-5) + T(n-6) + 21 = \cdots \end{aligned}$$

The expansion stops when the arguments reach 0 or 1. After two expansions the smallest argument is already $n-6$, and the number of terms keeps growing: we may suspect that $T(n)$ is exponential.

Proposition 3.1: The recursive algorithm is exponential

The running time of Algorithm 3.1 satisfies

$$T(n) = \Omega\big(2^{n/2}\big) = \Omega\big((\sqrt{2})^n\big).$$

Proof. T is non-decreasing: $T(1) = T(0)$, and for $n \geq 2$ we have $T(n) \geq T(n-1)$.

Replacing $T(n-1)$ with the smaller $T(n-2)$,

$$T(n) = T(n-1) + T(n-2) + 3 \geq 2\,T(n-2) + 3.$$

Applying the same inequality repeatedly,

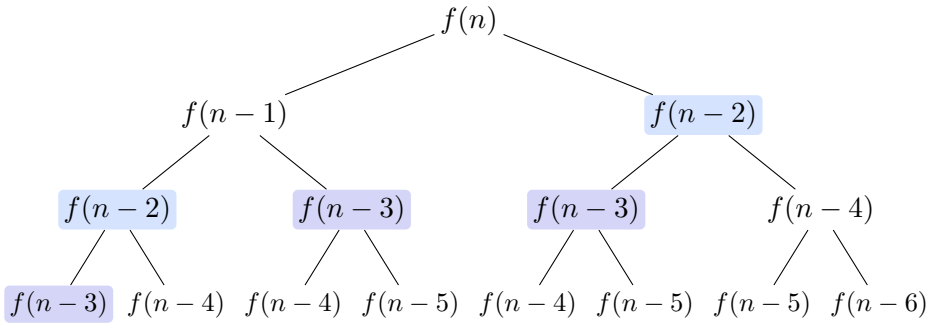
$$\begin{aligned}
T(n) &\geq 2\,\big(2T(n-4)+3\big) + 3 = 4\,T(n-4) + 9 \\
&\geq 4\,\big(2T(n-6)+3\big) + 9 = 8\,T(n-6) + 21 \\
&\geq 8\,\big(2T(n-8)+3\big) + 21 = 16T(n-8) + 45 \\
&\;\;\vdots \\
&\geq 2^i T(n-2i) + 3(2^i - 1) \geq 2^i T(n-2i) \qquad \text{after } i \text{ steps.}
\end{aligned}$$

For even n, take $i = n/2$:

$$T(n) \geq 2^{n/2}\,T(0) = 2 \cdot 2^{n/2} = 2(\sqrt{2})^n.$$

For odd n, take $i = (n-1)/2$ and use $T(1) = 2$ in the same way. □

Why is it so slow? To compute fib(n), the algorithm makes two calls, fib$(n-1)$ and fib$(n-2)$, each of which makes two more calls, and so on. It computes the same values a lot of times, again and again: the work roughly doubles at each level of the recursion tree.



Here $f(k)$ stands for a call to fib(k). Already in the first levels, $f(n-2)$ is computed twice and $f(n-3)$ three times.

3.1.2 Avoiding repeated work

There are two ways to fix the problem:

- **Store the values** (memoization): using some memory, we compute each fib(k) once and look it up afterwards.
- **Compute bottom-up**: start from the base cases and fill the table from left to right, $1+1=2$, $2+1=3$, $3+2=5$, $5+3=8$, …, each value being the sum of the previous two.

Algorithm 3.2: Iterative Fibonacci

```
1: function FibonacciIterative(n)
2:     if n ≤ 1 then
3:         return 1
4:     end if
5:     previous ← 1                                   ▷ fib(0)
6:     current ← 1                                    ▷ fib(1)
7:     for i = 2 to n do
8:         new ← previous + current
9:         previous ← current
10:        current ← new
11:    end for
12:    return current
13: end function
```

The loop runs $n-1$ times with a constant amount of work per iteration, so the running time is $T(n) = O(n)$: this algorithm is linear.

Note: Indexing and closed form

A common convention is $F_0 = 0$, $F_1 = 1$: with it, the iterative algorithm returns n for $n \leq 1$ and starts from 0 and 1. With that convention $\text{fib}(n) = F_{n+1}$, so the two sequences differ only by a shift. There is also a closed formula, which we do not prove:

$$F_n = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right], \qquad \text{fib}(n) = F_{n+1}.$$

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3.2 Binary search

Problem: Search

Input: an array A of n numbers and a value x.
Output: TRUE if x is in A, FALSE otherwise.

We could compare x with every element, with linear complexity. If A is *sorted*, we do not need to check all the elements: we look at the central element and compare it with x. If x is larger, it can only be in the right half; if smaller, only in the left half. We then repeat on that half.

Example 7: Searching a sorted array

Let $A = [1, 3, 7, 10, 13, 24, 30, 33, 39, 55]$ and $x = 39$.

1. The central element is $A[5] = 13 < 39$, so we keep the right part $[24, 30, 33, 39, 55]$.
2. Its central element is $33 < 39$, so we keep $[39, 55]$.
3. Its central element is $39 = x$: found.

Algorithm 3.3: Binary search

```
1: function BinarySearch(A, x)
2:     n ← len(A)
3:     if n = 0 then
4:         return FALSE
5:     end if
6:     if x < A[1] or x > A[n] then                ▷ x outside the range of A
7:         return FALSE
8:     end if
9:     mid ← ⌈n/2⌉
10:    if x = A[mid] then
11:        return TRUE
12:    else if x < A[mid] then
13:        return BinarySearch(A[1, ..., mid − 1], x)
14:    else
15:        return BinarySearch(A[mid + 1, ..., n], x)
16:    end if
17: end function
```

The *ceiling* $\lceil y \rceil$ is the smallest integer greater than or equal to y: for example, $\lceil 3 \rceil = 3$ and $\lceil 3.1 \rceil = 4$.

Example 8: An unsuccessful search

Let $A = [13, 17]$ and $x = 15$.

1. Since $13 < 15 < 17$, x lies within the range of A.
2. The central element is $A[1] = 13 < 15$ (here $mid = \lceil 2/2 \rceil = 1$), so we keep the right part $[17]$.
3. Now $15 < 17$, so x is outside the range of $[17]$: the algorithm returns FALSE.

3.2.1 Running time

Every line of Algorithm 3.3 takes constant time, except the recursive call. The call works on an array of half the size, so it takes $T(n/2)$ steps instead of $T(n)$:

$$T(n) = \begin{cases} c_1, & n \leq 1, \\ T(n/2) + c_2, & \text{otherwise}, \end{cases}$$

where c_1 and c_2 are constants. For $n \geq 2$,

$$T(n) = T\left(\tfrac{n}{2}\right) + c_2 = T\left(\tfrac{n}{4}\right) + 2c_2 = T\left(\tfrac{n}{8}\right) + 3c_2 = \cdots = T\left(\tfrac{n}{2^i}\right) + i\,c_2.$$

We stop when the array has size 1, that is, when $n/2^i = 1$, or $i = \log_2 n$. Then $T(n/2^i) = T(1) = c_1$ and

$$T(n) = c_1 + c_2 \log_2 n = \Theta(\log n).$$

Note: Subarrays

We treat $A[1, \ldots, mid - 1]$ as a view of the original array, described by its first and last index. Copying the subarray at every call would cost linear time and destroy the logarithmic bound.

We cannot do better than $\Theta(\log n)$, but we are not able to prove this yet. Splitting into three parts instead of two does not help asymptotically: the recurrence becomes $T(n) = T(n/3) + c_2'$, so $T(n) = O(\log_3 n)$, and $O(\log_3 n) = O(\log_2 n)$ because $\log_3 n = \log_2 n / \log_2 3$.

4

Complexity classes

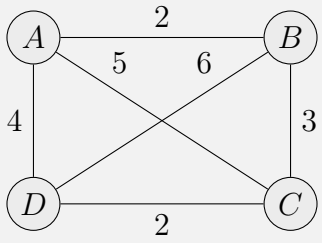
4.1 The traveling salesman problem

We want to visit all the nodes of a weighted graph, starting from a node and coming back to it, while minimizing the total cost. Every order of visit gives a tour, and different tours have different costs.

Problem: Traveling salesman problem (TSP)

Input: n cities and the cost of travelling between each pair of them.
Output: a tour that visits every city exactly once and returns to the starting city, with minimum total cost.

Example 9: Four cities



Starting from A, there are $3! = 6$ orders of visit, which give three distinct tours (each one can be travelled in two directions):

$$A\,B\,C\,D\,A: \ 2+3+2+4=11,$$
$$A\,B\,D\,C\,A: \ 2+6+2+5=15,$$
$$A\,C\,B\,D\,A: \ 5+3+6+4=18.$$

The optimal tour costs 11.

The brute-force approach tries all tours: with n cities there are $(n-1)!$ orders of visit, and

$$n! = 1 \cdot 2 \cdot 3 \cdots (n-1) \cdot n \geq 1 \cdot 2 \cdot 2 \cdots 2 = 2^{n-1},$$

so the factorial grows even faster than an exponential: this is not good at all. No known algorithm finds the shortest tour quickly.

4.2 Decision problems, P and NP

We usually study *decision problems*, whose answer is yes or no. An optimization problem such as minimizing costs or maximizing profits becomes a decision problem by fixing a threshold. For TSP: *is there a tour that visits all the nodes with total cost at most* 14*?* If the answer is yes, we can ask whether we can do better, for example cost at most 13.

Definition 4.1: The class P

P is the class of problems that we can solve in *polynomial time* (poly-time), that is, in time $O(n^k)$ for some constant k.

Definition 4.2: The class NP

NP is the class of problems for which, if someone gives us a candidate solution, we can verify in polynomial time whether the answer is YES.

Example 10: Easy to verify

- *Rubik's cube*: it is easy to check whether a cube is solved.
- *TSP*: given a tour, we add up its costs and check whether the total is at most 14.
- *Sudoku*: it is easy to check whether a filled grid is correct.

What these problems have in common is that, given a solution, we can verify in poly-time whether it is correct, even though finding one may be hard.

Max-Find and binary search are in P. Every problem in P is also in NP: if we can solve a problem quickly, we can verify an answer by solving it ourselves. Hence $\mathrm{P} \subseteq \mathrm{NP}$.

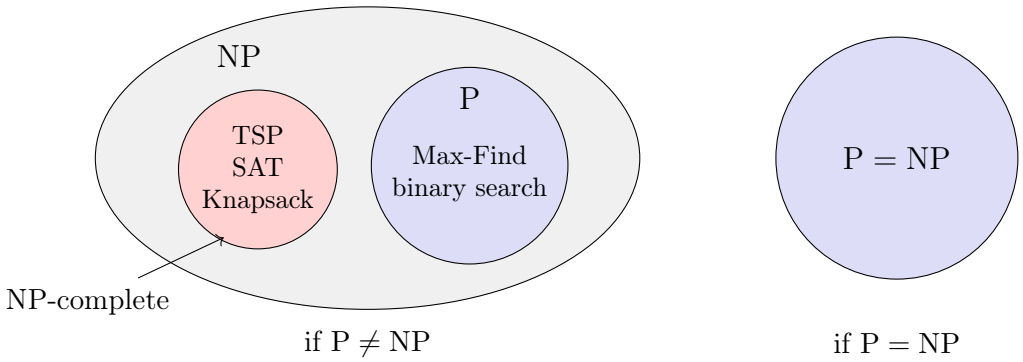
Note: What NP stands for

NP stands for *nondeterministic polynomial time*, not "non polynomial". A problem in NP may well be easy: every problem in P is in NP.

Are there problems in NP that are not in P? Nobody knows.

$$\textit{Is } \mathrm{P} = \mathrm{NP}\textit{, or is } \mathrm{P} \subsetneq \mathrm{NP}\textit{? We don't know.}$$

This is one of the most important open questions in computer science.



4.3 Reductions and NP-completeness

Some problems in NP are as hard as any other problem in NP.

Definition 4.3: NP-complete problem

A problem X is *NP-complete* if:

1. $X \in$ NP;
2. every problem $Y \in$ NP can be *reduced* in polynomial time to X.

A reduction of Y to X solves Y using an algorithm for X plus a polynomial amount of extra work.

Example 11: Max-Find and sorting

If we can sort an array in poly-time, we can also find its maximum in poly-time: sort it and take the last element. Max-Find reduces to sorting.

The converse also holds: if we can find the maximum, we put it in the last position and repeat on the remaining elements, until the whole array is sorted. This uses n calls to Max-Find, a polynomial amount of work, so sorting reduces to Max-Find.

Surprisingly, NP-complete problems exist, such as *SAT* (Boolean satisfiability), which we will not study in detail. If P = NP, the classes collapse: NP-complete problems would be solvable in polynomial time, like every other problem in NP.

Note: Are TSP, Sudoku and Rubik's cube NP-complete?

Complexity classes describe how the cost grows with the input size, so we must consider generalized versions of these puzzles:

- the decision version of TSP ("is there a tour of cost at most k?") is NP-complete;

- Sudoku on $n^2 \times n^2$ grids ("can this partially filled grid be completed?") is NP-complete;
- for the $n \times n \times n$ Rubik's cube, deciding whether a position can be solved in at most k moves is NP-complete (proved in 2018).

The classic 9×9 Sudoku and $3 \times 3 \times 3$ cube have a fixed size, so asymptotic statements do not apply to them directly.

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Corollary 4.1: Solving one NP-complete problem

Let X be NP-complete. If X can be solved in poly-time, then every problem in NP can be solved in poly-time. Equivalently, if some $Y \in \text{NP}$ cannot be solved in poly-time, then neither can X.

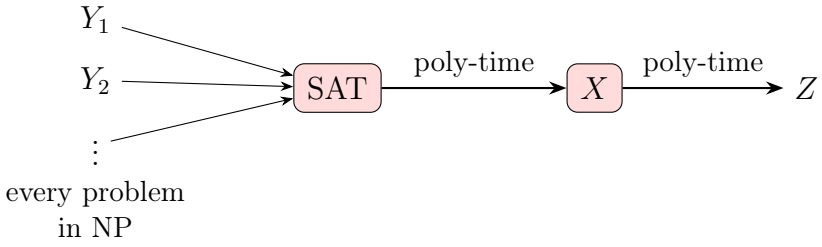
SAT was the first problem proved to be NP-complete: every problem in NP can be reduced to SAT (the Cook–Levin theorem). To show that a new problem is NP-complete, we no longer need to reduce every problem in NP to it.

Proposition 4.1: Proving NP-completeness

To prove that a problem X is NP-complete, show that:

1. $X \in \text{NP}$;
2. there is a problem Y already known to be NP-complete and a poly-time reduction of Y to X.

This works because reductions compose: every problem in NP reduces to Y, and Y reduces to X, so every problem in NP reduces to X. The composition of two polynomials is still a polynomial.



4.4 Some NP-complete problems

4.4.1 Knapsack

We want to fill a backpack with the most valuable set of items without exceeding its capacity.

Problem: Knapsack

Input: a set of items $I = \{1, 2, \ldots, n\}$, where item i has a weight w_i and a value v_i, and a budget B.
Output: a subset of items $S \subseteq I$ such that

$$\sum_{i \in S} w_i \leq B \qquad \text{that maximizes} \qquad \sum_{i \in S} v_i.$$

Example 12: A knapsack instance

Let $B = 55$ and

i	1	2	3	4	5	6	7
w_i	10	20	6	30	15	20	50
v_i	10	15	8	17	12	17	20

We must maximize $\sum_{i \in S} v_i$ subject to the constraint $\sum_{i \in S} w_i \leq 55$. One feasible solution is $S = \{1, 2, 3, 5\}$, with total weight 51 and total value 45. A solution taken "at random" like this one is probably not the best: in Example 16 we will find one of value 47.

4.4.2 Set cover

Suppose we want to rob a casino: we need people who can do different things.

Problem: Set cover

Input: a task that requires the skills $s_1, \ldots, s_n$, and m experts $E_1, \ldots, E_m$, where each $E_i \subseteq \{1, \ldots, n\}$ is the subset of skills that expert i has.
Output: the smallest number of experts that together cover all the skills.

In general, the skills are elements $e_1, \ldots, e_n$ and the experts are sets $S_1, \ldots, S_m$ of elements. We want the minimum number of sets whose union contains all the elements.

Example 13: Covering six skills

	s_1	s_2	s_3	s_4	s_5	s_6
E_1	×		×	×		
E_2	×	×				
E_3		×			×	
E_4	×			×		×
E_5			×			×

We must take E_3, because it is the only expert with skill s_5. E_3 covers s_2 and s_5; no single other expert covers all of s_1, s_3, s_4, s_6, so we need at least 3 experts. One optimal solution is $\{E_3, E_4, E_5\}$.

A heuristic method: take E_3, then repeatedly take the expert who covers the largest number of skills not yet covered. Such solutions are not optimal in general.

4.4.3 Subset sum

Problem: Subset sum

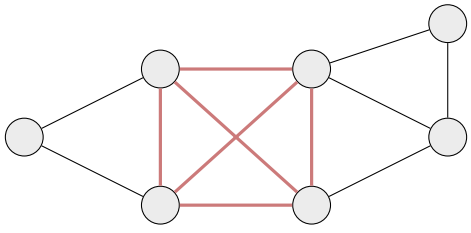
Input: a list of integers.
Output: a nonempty sublist whose elements sum to 0, if one exists.

For the list $[3, 7, -3, -8, 10, -16, 2, 9]$, the sublists $[3, -3]$ and $[7, 9, -16]$ both sum to 0.

4.4.4 Maximum clique

Problem: Maximum clique

Input: a graph.
Output: a largest *clique*, that is, a largest set of nodes that are all connected to one another.



The thick edges form a clique of size 4, the largest in this graph.

Proposition 4.2: Some NP-complete problems

The decision versions of Knapsack, Set cover, Subset sum and Max clique are NP-complete.

We do not prove this: each proof is a poly-time reduction from a known NP-complete problem (Proposition 4.1).

4.5 NP-hard problems

Definition 4.4: NP-hard problem

A problem X is *NP-hard* if every problem in NP can be reduced to X in polynomial time. Unlike an NP-complete problem, X does not have to belong to NP.

NP-hard problems are more general: NP-complete means NP-hard and in NP. For example, the optimization versions of Knapsack and TSP (find the best solution, not just answer yes or no) are NP-hard.

Note: Minimum spanning tree

The minimum spanning tree problem is *not* NP-hard (unless P = NP): Kruskal's and Prim's algorithms solve it in polynomial time. Its relative, the Steiner tree problem, which must connect only a given subset of the nodes, is NP-hard.

5 Approximation and heuristics

For NP-complete and NP-hard problems we do not know fast exact algorithms. Two ways out are approximation algorithms, which come with a guarantee, and heuristics, which do not.

5.1 Approximation algorithms

Definition 5.1: c-approximation algorithm

Let $c \geq 1$. For an input I, let OPT_I be the value of an optimal solution. A *c-approximation algorithm* returns, for every input I, a solution whose value is

- at most $c \cdot \mathrm{OPT}_I$, for a *minimization* problem;
- at least OPT_I / c, for a *maximization* problem.

Example 14: Reading the guarantee

- *TSP (minimization)*: suppose that for an input I the optimal tour costs $\mathrm{OPT}_I = 5000$. A 2-approximation algorithm guarantees a tour of cost at most 10 000: it may cost more than 5000, but we are sure that it does not exceed 10 000. For $c = 1.1$, the tour costs at most $1.1 \cdot 5000 = 5500$, no more than 10% above the optimum.
- *Knapsack (maximization)*: if $\mathrm{OPT}_I = 1000$, a 1.1-approximation guarantees a value of at least $1000/1.1 \approx 909$.

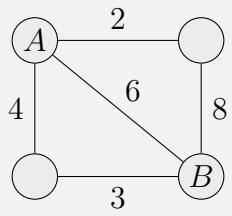
5.1.1 Guarantees without knowing the optimum

How can we prove that a solution is a c-approximation if we cannot find the optimal solution? We may not find OPT_I, but we can often find a *lower bound* $L \leq \mathrm{OPT}_I$ (for a

minimization problem). Then

$$\frac{\text{output}}{\text{OPT}_I} \leq \frac{\text{output}}{L}.$$

Example 15: Using a lower bound



Suppose every solution must travel from A to B. Every route from A to B costs at least 6, so $\text{OPT}_I \geq 6$: 6 is a lower bound. If our algorithm outputs a solution of cost 17, is it good?

$$\frac{\text{output}}{\text{OPT}_I} = \frac{17}{\text{OPT}_I} \leq \frac{17}{6} \approx 2.83,$$

so our solution is at most 17/6 times the optimum.

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5.1.2 How well can we approximate?

Different problems behave very differently:

- For *set cover*, where we minimize the number of sets, a simple greedy algorithm is, at worst, an $O(\log n)$-approximation.
- Many other problems cannot be approximated well in poly-time. For *max clique*, no polynomial-time algorithm can guarantee even a factor like $\sqrt{n}$, unless P = NP.

Some approximation algorithms are also very inefficient. This is why there is another approach: heuristics.

5.2 Heuristics

A *heuristic* is a method that usually finds good solutions quickly, but without any guarantee on their quality.

5.2.1 A greedy heuristic for knapsack

For each item, consider the value per unit of weight

$$r_i = \frac{v_i}{w_i}.$$

We start putting into the backpack the items with the largest r_i, and keep adding items as long as the constraint is satisfied. This usually gives a good solution, but not always the optimal one.

Algorithm 5.1: Greedy knapsack

sort the items so that $r_{i_1} \geq r_{i_2} \geq \cdots \geq r_{i_n}$
$S \leftarrow \varnothing$, $W \leftarrow 0$
for $k = 1$ **to** n **do**
    if $W + w_{i_k} \leq B$ **then**
        $S \leftarrow S \cup \{i_k\}$, $W \leftarrow W + w_{i_k}$
    end if
end for
return S

Example 16: Greedy on the knapsack instance

For the instance in Example 12,

$$r_1 = 1, \quad r_2 = \tfrac{3}{4}, \quad r_3 = \tfrac{4}{3}, \quad r_4 = \tfrac{17}{30}, \quad r_5 = \tfrac{4}{5}, \quad r_6 = \tfrac{17}{20}, \quad r_7 = \tfrac{2}{5}.$$

Item 3 has the largest ratio, so it is the first one we add. The order is $3, 1, 6, 5, 2, 4, 7$:

- add 3 (weight 6), 1 (weight 16), 6 (weight 36) and 5 (weight 51);
- items 2, 4 and 7 no longer fit ($51 + 20 > 55$).

The result is $S = \{1, 3, 5, 6\}$, with weight 51 and value 47. Trying all subsets shows that 47 is optimal for this instance.

The greedy solution is not optimal in general.

Example 17: When greedy fails

Let $B = 100$, ten items with $w_1 = \cdots = w_{10} = 50$ and $v_1 = \cdots = v_{10} = 50$ (each with ratio 1), and an item 0 with $w_0 = 51$ and $v_0 = 52$, whose ratio is slightly larger than 1. Greedy takes item 0 first, and then no other item fits ($51 + 50 > 100$): the value is 52. Taking two items of weight 50 gives value 100.

No known algorithm solves knapsack in poly-time: it is NP-complete. If we could "break" the items and take a fraction of one of them (the *fractional* knapsack), the greedy algorithm would be perfect: we fill the remaining capacity with a fraction of the next item.

5.2.2 Greedy algorithms and meta-heuristics

Definition 5.2: Greedy algorithm

A *greedy* algorithm makes, at each step, the choice that is best for now, hoping that it will also work well in the future, and continues in the same way.

The knapsack heuristic above and the set cover heuristic are both greedy algorithms. Greediness is an example of a *meta-heuristic*: a general strategy, a wider class that includes many heuristics. Another example is *local search*: starting from a solution, such as a tour, we try small modifications (for example, exchanging two of its edges for two others) and keep those that improve it, searching for the optimal one.

6

Distance measures

6.1 Why distances?

Many problems ask us to find distances or similarities between objects. For example, given some documents and a query, we want the documents most similar to the query; or we want groups of similar documents, a *cluster*. The input is a set of objects and the output a group of them. To do this we use different kinds of distances: a function that measures the distance between two elements defines a relation between them.

6.2 Distance functions

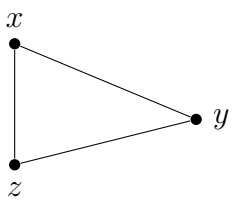
Definition 6.1: Distance function

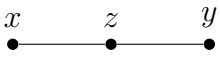
Let U be a universe of elements (documents, objects, strings, ...). A *distance function* is a map

$$d : U \times U \longrightarrow \mathbb{R},$$

taking two elements and returning a number, which satisfies, for all $x, y, z \in U$:

1. $d(x, y) \geq 0$, and $d(x, y) = 0$ if and only if $x = y$;
2. $d(x, y) = d(y, x)$ (symmetry);
3. $d(x, y) \leq d(x, z) + d(z, y)$ (triangle inequality).





equality when z lies between

We should define distances that make sense for the problem: what does it mean for two objects to be "close"? It depends on the problem we are considering.

6.3 L_p distances

We have defined what a distance function is (Definition 6.1). We now look at concrete distances, starting with objects described by vectors of numbers, that is, $U = \mathbb{R}^m$.

Suppose we are designing an app like Google Maps that shows the restaurants near us. On the map, our position and each restaurant are points with two coordinates, that is, vectors in $\mathbb{R}^2$. To decide which restaurants are near, we need a distance between two points. The natural choice is the length of the straight segment that joins them: the Euclidean distance. The same formula works for vectors with any number m of coordinates.

Definition 6.2: Euclidean distance

For $U = \mathbb{R}^m$, $\vec{x} = [x_1, \ldots, x_m]$ and $\vec{y} = [y_1, \ldots, y_m]$, the *Euclidean distance*, or L_2 distance, is

$$d(\vec{x}, \vec{y}) = \sqrt{\sum_{j=1}^{m} (x_j - y_j)^2}.$$

It is clearly non-negative and symmetric; it also satisfies the triangle inequality, but we do not prove it.

Definition 6.3: L_p distance

For a constant $p \geq 1$,

$$d_p(\vec{x}, \vec{y}) = \left(\sum_{j=1}^{m} |x_j - y_j|^p \right)^{1/p} = \|\vec{x} - \vec{y}\|_p.$$

Changing p gives different distances, which weigh the differences between coordinates differently:

- $p = 2$: the Euclidean distance.
- $p = 1$: the *Manhattan distance*, $\sum_j |x_j - y_j|$.
- $p \to \infty$: the L_∞ distance, $d_\infty(\vec{x}, \vec{y}) = \max_{i \in \{1,\ldots,m\}} |x_i - y_i|$.
- "L_0": the number of coordinates in which the vectors differ, $|\{i : x_i \neq y_i\}|$. For binary strings this is the *Hamming distance*.

Example 18: Counting the coordinates that differ

- The binary strings 00110 and 10100 differ in 2 positions (the first and the fourth), so their Hamming distance is 2.
- The vectors $[3, 1, 3]$ and $[0, 1, 3]$ differ only in the first coordinate, so their L_0 distance is 1.

Example 19: Comparing L_p distances

Let

$$\vec{x} = [3, 7, 0, 2, 1], \qquad \vec{y} = [4, 5, 10, 7, 3], \qquad \vec{z} = [5, 2, 5, 6, 5].$$

The coordinate differences are $|\vec{x} - \vec{y}| = [1, 2, 10, 5, 2]$ and $|\vec{x} - \vec{z}| = [2, 5, 5, 4, 4]$.

p	$\|\vec{x} - \vec{y}\|_p$	$\|\vec{x} - \vec{z}\|_p$
1	$1 + 2 + 10 + 5 + 2 = 20$	$2 + 5 + 5 + 4 + 4 = 20$
2	$(1 + 4 + 100 + 25 + 4)^{1/2} \approx 11.58$	$(4 + 25 + 25 + 16 + 16)^{1/2} \approx 9.27$
3	$(1 + 8 + 1000 + 125 + 8)^{1/3} \approx 10.45$	$(8 + 125 + 125 + 64 + 64)^{1/3} \approx 7.28$
∞	10	5

With $p = 1$ the two distances are equal. As p increases, both distances decrease, and the larger differences weigh more: $\vec{y}$ differs from $\vec{x}$ by 10 in one coordinate, so it ends up farther than $\vec{z}$, whose differences are moderate everywhere.

6.4 Distances between sets

So far we have worked with vectors. Now imagine that we want to compare the customers of a supermarket based on what they bought.

Example 20: Shopping baskets

Each letter is a product: the first person bought apples (A), bananas (B), carrots (C) and donuts (D):

$$P_1 = \{A, B, C, D\}, \qquad P_2 = \{A, E, G, H\}, \qquad P_3 = \{B, C, E, F\}.$$

Intuitively, persons 1 and 3 are more similar than persons 1 and 2.

6.4.1 Sets as binary vectors

We can write each set as a binary vector, with one coordinate per product, and use the L_1 distance:

	A	B	C	D	E	F	G	H
P_1	1	1	1	1	0	0	0	0
P_2	1	0	0	0	1	0	1	1
P_3	0	1	1	0	1	1	0	0

Then $|P_1 - P_2| = [0, 1, 1, 1, 1, 0, 1, 1]$ and $|P_1 - P_3| = [1, 0, 0, 1, 1, 1, 0, 0]$, so

$$\|P_1 - P_2\|_1 = 6, \qquad \|P_1 - P_3\|_1 = 4.$$

So P_1 is closer to P_3, as we expected. In general,

$$\|P - Q\|_1 = |P| + |Q| - 2|P \cap Q| = |P \triangle Q|,$$

the size of the *symmetric difference* $P \triangle Q$, the elements in exactly one of the two sets.

6.4.2 Counting common items

The formula above depends on the intersection $P \cap Q$. Let us look at the elements in common directly:

$$|P_1 \cap P_2| = 1, \qquad |P_1 \cap P_3| = 2.$$

Is this a good distance? We should check the three properties. Symmetry certainly holds, but the larger the value, the *closer* the two sets: this is a *similarity*, not a distance. It is often more natural to talk about similarity than about distance.

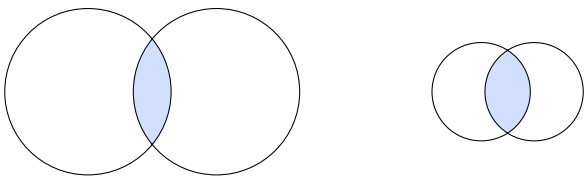
Definition 6.4: Similarity function

A *similarity function* is a map $s : U \times U \to \mathbb{R}$ such that $s(x, y) = s(y, x)$ for all $x, y \in U$, and whose value is larger the more similar x and y are. Similarities are often normalized to take values in $[0, 1]$, with $s(x, x) = 1$ for every x.

Unlike distances, similarities have no standard list of properties: the ones above are the usual conventions. The two notions go in opposite directions: a large similarity means a small distance. Here we do not see a "natural" distance function, and we cannot simply take the negative of the similarity, because distances cannot be negative.

6.4.3 Normalizing the intersection

The size of the intersection alone, however, is misleading. Two pairs of sets can have intersections of the same size, but if the sets in the second pair are smaller, the intersection is a larger fraction of them, so they are more similar.



1 2

Same intersection, different set sizes: the sets in pair 2 are more similar, because their intersection is a larger fraction of them.

We can normalize the intersection in several ways:

$$\frac{|P \cap Q|}{\max\{|P|, |Q|\}}, \qquad \frac{|P \cap Q|}{|P \cup Q|}, \qquad \frac{|P \cap Q|}{|P| + |Q|}.$$

Definition 6.5: Jaccard similarity

The *Jaccard similarity* of two finite sets P and Q is

$$J(P, Q) = \frac{|P \cap Q|}{|P \cup Q|},$$

where $|\cdot|$ is the cardinality (not the absolute value).

The Jaccard similarity is symmetric, lies between 0 and 1, and gives $J(P, P) = 1$ for every nonempty P: it is a normalized similarity function (Definition 6.4). For the baskets in Example 20,

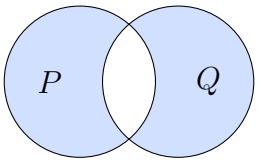
$$J(P_1, P_2) = \frac{|\{A\}|}{|\{A, B, C, D, E, G, H\}|} = \frac{1}{7}, \qquad J(P_1, P_3) = \frac{|\{B, C\}|}{|\{A, B, C, D, E, F\}|} = \frac{2}{6} = \frac{1}{3},$$

so P_1 is more similar to P_3, as expected.

Definition 6.6: Jaccard distance

We turn the Jaccard similarity into a distance by taking

$$d_J(P, Q) = 1 - \frac{|P \cap Q|}{|P \cup Q|} = \frac{|P \cup Q| - |P \cap Q|}{|P \cup Q|} = \frac{|P \triangle Q|}{|P \cup Q|} = \frac{|P \setminus Q| + |Q \setminus P|}{|P \cup Q|}.$$



The symmetric difference $P \triangle Q$: everything except the intersection.

One can prove that the Jaccard distance d_J is a distance in the sense of Definition 6.1. For the baskets, $d_J(P_1, P_2) = 6/7$ and $d_J(P_1, P_3) = 4/6 = 2/3$.

7

Distances between strings

Start of lecture 12 October

7.1 Why compare strings?

Biological data are sequences of bases such as `CTAGGATCGGAC`, and DNA sequences have particular properties: we may want to know how "similar" two parts of DNA are. Search engines need similarity between strings too: if we type "Sceince" instead of "Science" on Google, it still understands what we mean.

Example 21: Which strings are closer?

$$X = \texttt{CTAGGAT}, \qquad Y = \texttt{CCAGAGT}, \qquad Z = \texttt{AGCCCTA}.$$

Which pair is more similar, X and Y, or X and Z? We would say X and Y. The answer is intuitive, but which property are we using? How do we define a "distance measure" between strings?

7.2 First attempts

- **Compare letter by letter.** We could compare the strings position by position, as with the Hamming or L_1 distance. But what if the strings have different lengths?
- **Letters as numbers.** Another idea that might come to mind is to associate each letter with its position in the alphabet ($A = 1$, $C = 3$, $G = 7$, $T = 20$) and to compute the L_1 distance between the resulting vectors. For example,

`CAGT` :	3	1	7	20
`ACTT` :	1	3	20	20
difference :	2	2	13	0

$$\Longrightarrow \qquad 2 + 2 + 13 + 0 = 17.$$

The larger the number, the farther the strings, and the distance is symmetric. However, it does not work well:

  - `SAPIENZA` and `SSAPIENZA`, typed with an extra `S` by mistake, are almost identical, but the extra letter shifts all the following ones and the distance explodes;
  - the distance between A and T is larger than between A and B only because of the alphabetical order.

- **Count the characters.** We could count how many times each character appears. Biologically, however, `AAGGGTT` and `TGAAGTG` are very different, even though they contain exactly the same characters.
- **Look at common pieces.** `CATGAGG` and `CAGAGT` share `CA` and `GAG`, even though they have different lengths.
- **Count modifications.** We could take the second string and count how many changes are needed to make it equal to the first. For `SAPIENZA` and `SAAPIEMZA`, we delete the extra `A` and replace `M` with `N`: two changes, so the distance is 2.

The last two ideas are the promising ones: common pieces lead to subsequences, and modifications lead to the edit distance. Consider sequences over an alphabet.

7.3 Subsequences

Definition 7.1: Subsequence

A *subsequence* of a sequence S is a sequence that contains a subset of the characters of S in the same order, not necessarily consecutive.

Example 22: Subsequences

For $S =$ `ATCGGACTGA`, all of the following are subsequences: `ATCGGACTGA` itself, `ACGGCTA`, `AAA`, and the empty string.

Definition 7.2: Common subsequence

Given two sequences S and T, a *common subsequence* is a sequence that is a subsequence of both S and T.

For $S =$ `ATGCATG` and $T =$ `AGACTAG`, `AAT` and `AGAG` are common subsequences, and so is the empty string.

7.3.1 The longest common subsequence

Definition 7.3: Longest common subsequence (LCS)

The *longest common subsequence* of S and T is a common subsequence of maximum length. We write $\mathrm{LCS}(S, T)$ for its length.

We can define the similarity of two strings as the length of their LCS.

Example 23: Longest common subsequences

- For `ATGA` and `TAG`, both `AG` and `TG` are longest common subsequences: the LCS need not be unique, and here none is longer than 2.
- Compare `SAPEMZA` with `SAPIENZA` and with `SALERNO`:

  $\mathrm{LCS}(\texttt{SAPIENZA}, \texttt{SAPEMZA}) = 6$ (`SAPEZA`), $\mathrm{LCS}(\texttt{SAPEMZA}, \texttt{SALERNO}) = 3$ (`SAE`).

  `SAPEMZA` has more in common with `SAPIENZA` than with `SALERNO`.
- In Example 21, $\mathrm{LCS}(X, Y) = 5$ (for example `CAGGT`) while $\mathrm{LCS}(X, Z) = 3$ (`CTA`), which confirms our intuition.
- For `CAGTGACCGATAC` and `GACCAGTCAGCAC` it is already hard to find the LCS by hand. Its length is 8, for example `CAGTGCAC`.

Checking all subsequences is not an option: a string of length k has up to 2^k subsequences, one for each subset of its positions, so brute force needs about $2^{\min\{|S|,|T|\}}$ checks.

7.4 Edit distance

How do we compare two strings? Option 1: transform one string into the other. Option 2: look at their subsequences. The two options turn out to be closely related.

Definition 7.4: Edit distance

Given two sequences S and T, the *edit distance* $\mathrm{ED}(S, T)$ is the minimum number of insertions and deletions of characters needed to transform S into T.

Example 24: Transforming a string

Let $S =$ `CATGA` and $T =$ `ACTGC`:

$$\texttt{CATGA} \xrightarrow{\text{delete C}} \texttt{ATGA} \xrightarrow{\text{insert C}} \texttt{ACTGA} \xrightarrow{\text{insert C}} \texttt{ACTGAC} \xrightarrow{\text{delete A}} \texttt{ACTGC}.$$

Four changes suffice, and in fact $\mathrm{ED}(S, T) = 4$.

We could also delete all the letters one by one and insert the new ones, but that is not the shortest way. When the strings are longer, it is hard to see the minimum number of steps.

The edit distance is symmetric, $\mathrm{ED}(S,T) = \mathrm{ED}(T,S)$, because we can perform all the steps in reverse: each insertion becomes a deletion and vice versa.

Note: Substitutions

Some definitions also allow *substitutions*, replacing one character with another at cost 1. With insertions and deletions only, a substitution costs 2: one deletion and one insertion.

7.4.1 Edit distance and LCS

A natural strategy for transforming S into T: find an LCS, delete everything else from S, leaving only the LCS, and then insert the characters of T that are missing.

Proposition 7.1: Edit distance from the LCS

$$\mathrm{ED}(S,T) = \underbrace{\big(|S| - \mathrm{LCS}(S,T)\big)}_{\text{characters to delete}} + \underbrace{\big(|T| - \mathrm{LCS}(S,T)\big)}_{\text{characters to insert}} = |S| + |T| - 2\,\mathrm{LCS}(S,T).$$

Why it holds. The strategy above uses exactly $|S| + |T| - 2\,\mathrm{LCS}(S,T)$ operations, so $\mathrm{ED}(S,T)$ is at most this number. Conversely, in any sequence of insertions and deletions, the characters of S that are never deleted appear in T in the same order: they form a common subsequence, of some length $k \le \mathrm{LCS}(S,T)$. We must then delete at least $|S| - k$ characters and insert at least $|T| - k$, so at least $|S| + |T| - 2k \ge |S| + |T| - 2\,\mathrm{LCS}(S,T)$ operations are needed. □

For Example 24, $\mathrm{LCS}(\texttt{CATGA}, \texttt{ACTGC}) = 3$ (for example `CTG`), so $\mathrm{ED} = 5 + 5 - 2 \cdot 3 = 4$.

7.5 A recursive formula for the edit distance

The optimal algorithm for computing the edit distance is not obvious. We now write a recursive formula, the starting point of *dynamic programming*.

Note: Notation

Consider two strings S and T with $|S| = n$ and $|T| = m$:

$$S = [S[1], S[2], \ldots, S[n]], \qquad T = [T[1], \ldots, T[m]].$$

We write $S^i = [S[1], \ldots, S[i]]$ and $T^j = [T[1], \ldots, T[j]]$ for the *prefixes* of length i and j.

Proposition 7.2: Recurrence for the edit distance

$$\mathrm{ED}(S^i, T^j) = \begin{cases} j, & \text{if } i = 0, \\ i, & \text{if } j = 0, \\ \mathrm{ED}(S^{i-1}, T^{j-1}), & \text{if } S[i] = T[j], \\ \min\{\mathrm{ED}(S^i, T^{j-1}),\ \mathrm{ED}(S^{i-1}, T^j)\} + 1, & \text{if } S[i] \neq T[j]. \end{cases}$$

- If one string is empty, we must insert (or delete) all the characters of the other.
- If the last characters are equal, we do not need to modify them: we can ignore them and consider the distance between S^{i-1} and T^{j-1}. For example,

$$\mathrm{ED}(\texttt{CATGCAT T}, \texttt{ACCTAGAC T}) = \mathrm{ED}(\texttt{CATGCAT}, \texttt{ACCTAGAC}).$$

- If the last characters differ, one of them must go: either we remove the last character of S, giving $\mathrm{ED}(S^{i-1}, T^j) + 1$, or we remove the last character of T (equivalently, we insert it at the end of S), giving $\mathrm{ED}(S^i, T^{j-1}) + 1$. For `CATGCAT T` and `ACCTAGAC A`, we remove either the `T` or the `A`. We take the better of the two options.

Example 25: Small edit distances

- $\mathrm{ED}(\texttt{TGA}, \texttt{TTA}) = 2$: we must remove `G` and add `T`. Also $\mathrm{ED}(\texttt{TG}, \texttt{TT}) = 2$, because we have to remove a character and then add one.
- For `CAG` and `TGA`, the last characters differ. Either we remove `G` from `CAG`, which leaves `CA` and `TGA` (+1); or we remove `A` from `TGA`, which leaves `CAG` and `TG` (+1), whose last characters are now equal. Both options lead to $\mathrm{ED}(\texttt{CAG}, \texttt{TGA}) = 4$.

Algorithm 7.1: Recursive edit distance

```
1: function ED(S^i, T^j)
2:     if i = 0 then
3:         return j
4:     else if j = 0 then
5:         return i
6:     else if S[i] = T[j] then
7:         return ED(S^{i-1}, T^{j-1})
8:     else
9:         return 1 + min{ED(S^i, T^{j-1}), ED(S^{i-1}, T^j)}
10:    end if
11: end function
```

Written directly as a recursive algorithm, this formula has an exponential running time.

Note: The same problem as Fibonacci

As with the recursive Fibonacci algorithm (Proposition 3.1), the recursion solves the same subproblems again and again. There are only $(n+1)(m+1)$ different subproblems $\mathrm{ED}(S^i, T^j)$, so storing each value once (dynamic programming) avoids the repeated work.

8

Memory and data representation

Start of lecture 19 October

8.1 Bytes, addresses and consecutive memory locations

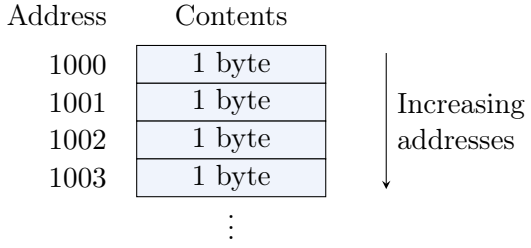
What is memory for? While a program runs, it needs somewhere to keep the data it is working with: numbers, text, lists and the intermediate results of calculations. Main memory (RAM) provides this working space, so the processor can read the data, perform operations and store the results for later steps of the program.

For example, after executing `x = 5`, Python must keep the object representing 5 available. A later instruction such as `print(x)` can retrieve it, and `x + 2` can use it to compute 7. Understanding how data are arranged and reached in memory will help us explain the cost of operations such as accessing an element of a list.

Definition 8.1: Bit and byte

A *bit* is a binary digit, either 0 or 1. A *byte* consists of eight bits.

Imagine memory as numbered cells, drawn vertically below. Each cell holds one byte. The number on the left is its address: it identifies the cell, while the contents are stored inside it.



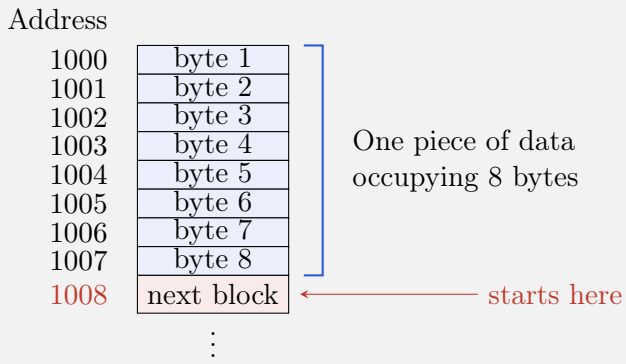
The address and the contents are different things: the label 1000 tells us where the first displayed byte is, not what value it holds. We use illustrative decimal addresses throughout these drawings.

Definition 8.2: Memory address

In byte-addressable memory, a *memory address* identifies the location of one byte. The address specifies *where* data are stored; the contents are the data stored at that location.

Example 26: A block occupying eight consecutive bytes

Suppose a piece of data occupies eight bytes starting at address 1000. Follow the addresses down the drawing:



The starting byte is already the first of the eight, so the last byte is at $1000 + 7 = 1007$. The first address *after* the block is

$$1000 + 8 = \boxed{1008}.$$

If a second eight-byte block starts immediately afterwards, its starting address is 1008.

This explains why a memory diagram can show addresses advancing in blocks: it may label the start of each block rather than drawing every individual byte. For eight-byte blocks placed consecutively, the starting addresses would be $1000, 1008, 1016, \ldots$.

8.2 Python names refer to objects

Now consider the Python assignment `x = 5`. We distinguish the name `x`, the object representing the integer 5, and the address where that object is stored.

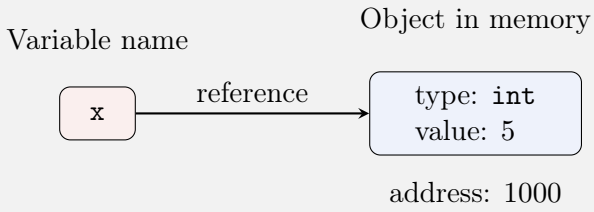
Definition 8.3: Reference to an object

A Python variable name is associated with an object through a *reference*. In our diagrams, an arrow from the name to the object represents that reference.

Example 27: Following the reference in `x = 5`

Step 1: an object represents the integer 5. The drawing shows its type, `int`, and its value, 5.

Step 2: the name `x` is associated with that object. The arrow lets us follow the reference from the name to the object.



The address 1000 is illustrative. It identifies the object's location, while 5 is the object's value and `x` is its name in this example.

Step 3: evaluate `print(x)`. Python looks up the object associated with `x`. Following the arrow reaches the integer object with value 5, so the output is `5`.

9 Text mining

Start of lecture 20 October

9.1 From raw text to terms

A search engine receives a query and looks for the top k most related documents in a collection, or *corpus*. We will turn this into a nearest-neighbour problem, but first we need to turn each document's text into terms that we can compare and count.

Definition 9.1: Text preprocessing

Text preprocessing transforms a document, initially a sequence of characters, into a sequence of words or terms suitable for the analysis that follows.



1. Tokenization. Split the text into units called *tokens*. For example, the document

"Today is Monday and the weather is awesome."

can be split into the word tokens

`Today`, `is`, `Monday`, `and`, `the`, `weather`, `is`, `awesome`.

The input was a string of characters; the output is now a sequence of separate terms.

2. Stopword removal. Remove words on a chosen *stopword list*, such as `the`, `is`, `to` and `and`. These words are often too common to help distinguish document topics. Depending

on the list, a phrase such as "to be or not to be" can lose most of its words. We will later examine how common words affect cosine similarity.

3. Text normalization. Map different written forms to a common form when they should match. For example,

$$\texttt{U.S.A.} \longrightarrow \texttt{USA}, \qquad \texttt{Monday} \longrightarrow \texttt{monday}.$$

The first transformation removes punctuation within an abbreviation; the second ignores capitalization. Other choices concern accents and hyphens. This normalization acts on the text; the vector normalization introduced later rescales numerical coordinates.

4. Stemming. Reduce related word forms to a common *stem*, using rules that remove affixes such as suffixes. For example,

$$\texttt{playing}, \texttt{plays}, \texttt{play} \quad \longrightarrow \quad \texttt{play}.$$

The aim is to count these forms together instead of treating each as a completely separate term. `NLTK` is a Python toolkit for text-processing operations.

9.2 From documents to vectors

How can we turn a text into something on which we can perform calculations? Suppose we want to find documents about similar topics. Before choosing a distance or a similarity function, we need a numerical representation of each document.

Definition 9.2: Corpus and vocabulary

A *corpus* is a collection of documents. Its *vocabulary* is the set of distinct terms appearing in those documents, after any preprocessing. We fix an order for the vocabulary and use the same order for every document.

Example 28: Counting words in two documents

Consider the documents

$$D_1: \quad \text{"today we started at three"},$$
$$D_2: \quad \text{"we played three by three"}.$$

For this example, we keep all the words. In the vocabulary order

$$(\texttt{at}, \texttt{by}, \texttt{played}, \texttt{started}, \texttt{three}, \texttt{today}, \texttt{we}),$$

their word counts are

	at	by	played	started	three	today	we
D_1	1	0	0	1	1	1	1
D_2	0	1	1	0	2	0	1

Each entry counts how many times a term occurs in a document. For example, the entry for `three` in D_2 is 2, because that word appears twice. Reading the rows gives the vectors

$$\mathbf{d}_1 = (1, 0, 0, 1, 1, 1, 1), \qquad \mathbf{d}_2 = (0, 1, 1, 0, 2, 0, 1).$$

The common coordinate order is essential: a coordinate must refer to the same word in both documents for a comparison to make sense.

Definition 9.3: Count-vector representation

Let the ordered vocabulary be $(t_1, \ldots, t_m)$. We represent document D_i by the vector $\mathbf{d}_i \in \mathbb{R}^m$, where

$$d_{ij} = \text{number of occurrences of term } t_j \text{ in document } D_i.$$

Thus every term defines one coordinate, and every document becomes a point in the same m-dimensional space. This is a basic *vector space model* of documents.

This representation records which terms occur and how often, but it does not preserve their order in the text. Consequently, identical count vectors do not necessarily imply identical meanings.

9.3 Why document length can distort a comparison

Now that documents are points, we can compare them using the L_2 distance already introduced in Definition 6.2. For a two-dimensional illustration, keep only the coordinates (`three`, `we`) from Example 28:

$$\mathbf{u}_1 = (1, 1), \qquad \mathbf{u}_2 = (2, 1).$$

These are two-coordinate views of the documents, rather than their complete vectors. Their distance is

$$\|\mathbf{u}_1 - \mathbf{u}_2\|_2 = \sqrt{(1-2)^2 + (1-1)^2} = 1.$$

We would like closeness between points to reflect similarity between documents. However, raw word counts introduce a difficulty.

Example 29: Repeating a document ten times

Here D_1 is the original five-word document:

$$D_1 : \quad \text{“today we started at three”}.$$

Construct D_3 by repeating this *entire sentence ten times inside a single document*:

$$D_3 : \quad \underbrace{\text{“today we started at three”} \cdots \text{“today we started at three”}}_{\text{10 copies of the sentence, 50 words in total}}.$$

Using the same vocabulary order as before, compare the counts:

	at	by	played	started	three	today	we
D_1	1	0	0	1	1	1	1
D_3	10	0	0	10	10	10	10

Thus D_3 is one longer document, not a collection of ten separate documents. Every term now occurs ten times as often:

$$\mathbf{d}_3 = 10\mathbf{d}_1.$$

In particular, `three` occurs ten times and `we` occurs ten times, so the two-coordinate view becomes $\mathbf{u}_3 = (10, 10)$. Its distance from the original is

$$\|\mathbf{u}_1 - \mathbf{u}_3\|_2 = \sqrt{(1-10)^2 + (1-10)^2} = 9\sqrt{2}.$$

Yet D_3 repeats exactly the same content: it adds no new information and does not change the topic. Every word retains the same proportion of the total text. The large distance is caused by the increase in length, rather than by a change in word composition.

9.4 Removing the effect of length: normalization

The idea is to rescale the counts so that multiplying every count by the same factor does not change the representation. A first way to do this is to divide each count by the total number of terms in the document: we then compare *relative frequencies*.

Example 30: From counts to proportions

As in Example 29, the documents are

$$D_1: \quad \text{“today we started at three”},$$

$$D_3: \quad \underbrace{\text{“today we started at three”} \cdots \text{“today we started at three”}}_{\text{10 copies of the sentence, 50 words in total}}.$$

	at	by	played	started	three	today	we
D_1	1	0	0	1	1	1	1
D_3	10	0	0	10	10	10	10

Document D_1 contains five words and D_3 contains fifty. For the coordinates (`three`, `we`), dividing by the respective document lengths gives

$$D_1: \quad \left(\frac{1}{5}, \frac{1}{5}\right) = (0.2, 0.2), \qquad D_3: \quad \left(\frac{10}{50}, \frac{10}{50}\right) = (0.2, 0.2).$$

The representations now agree. Notice that the denominators count *all* the terms in each document, not just the two displayed coordinates.

More generally, we can divide a nonzero document vector by its norm.

Definition 9.4: Document-vector normalization

For a nonzero vector $\mathbf{d}$ and a chosen norm, its normalized version is

$$\widehat{\mathbf{d}} = \frac{\mathbf{d}}{\|\mathbf{d}\|}.$$

Each coordinate is divided by the same positive number, and the resulting vector has norm 1.

For a complete vector of nonnegative term counts, the L_1 norm is the sum of all counts, hence the total number of terms counted in the document. Normalizing with L_1 therefore gives the relative frequencies used above. Normalizing with L_2 instead gives a vector of Euclidean norm 1; these are two different choices of scale.

Both choices remove a uniform multiplication of the counts. Indeed, for $c > 0$ and $\mathbf{d} \neq \mathbf{0}$,

$$\frac{c\mathbf{d}}{\|c\mathbf{d}\|} = \frac{c\mathbf{d}}{c\|\mathbf{d}\|} = \frac{\mathbf{d}}{\|\mathbf{d}\|}.$$

Thus repeating D_1 ten times yields the same normalized vector. Geometrically, normalization rescales the vector without changing its direction. A document with no counted terms has the zero vector and cannot be normalized by this formula.

9.5 Comparing directions: cosine similarity

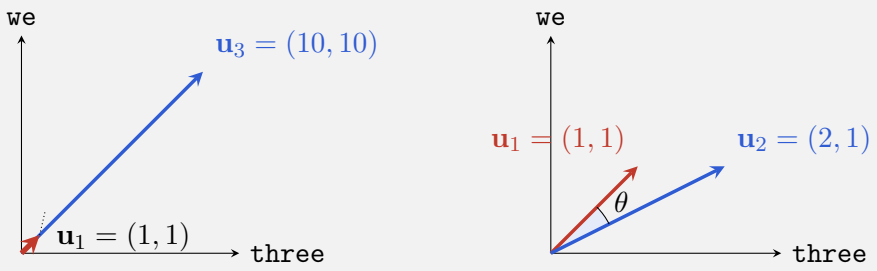
Normalization suggests a geometric way to compare documents: two vectors can have different lengths while pointing in the same direction. Now, we can consider another idea: compare the *angle* between the vectors. If their counts are proportional, their relative word composition is the same and the angle is zero.

Example 31: Same direction, different lengths

As before, keep only the coordinates (`three`, `we`):

$$\mathbf{u}_1 = (1, 1), \qquad \mathbf{u}_3 = (10, 10).$$

Draw each vector as an arrow starting at the origin. The arrow for $\mathbf{u}_3$ is ten times as long, but points in exactly the same direction as $\mathbf{u}_1$: the angle between them is $0°$. Now consider $\mathbf{u}_2 = (2, 1)$. Here `three` occurs twice as often as `we`, so its arrow points more towards the `three` axis and forms a positive angle with $\mathbf{u}_1$.



The smaller the angle, the more similar the proportions of terms in the two vectors. We turn this idea into a numerical similarity by taking the cosine of that angle.

Definition 9.5: Cosine similarity

For two nonzero document vectors $\mathbf{u}$ and $\mathbf{v}$, let θ be the angle between them. Their *cosine similarity* is

$$\mathrm{sim}_{\cos}(\mathbf{u}, \mathbf{v}) = \cos\theta.$$

For nonnegative count vectors, the angle lies between 0° and 90°, so the similarity lies between 0 and 1:

- At 0°, the similarity is 1: the vectors point in the same direction, and their counts are proportional.
- Between 0° and 90°, the similarity lies strictly between 1 and 0. As the angle increases, the similarity decreases.
- At 90°, the similarity is 0: there is no coordinate with a positive count in both vectors. For complete document vectors, this means that the documents share no counted terms.

In particular, D_1 and its tenfold repetition D_3 have cosine similarity 1. This measures similarity of word composition; as with the count representation itself, it does not capture word order or guarantee identical meanings. A zero vector has no direction, so the similarity above is not defined for an empty count vector.

9.5.1 Computing cosine similarity from coordinates

We do not need to measure the angle first. For two nonzero vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^m$, we can compute its cosine directly from their coordinates:

$$\mathrm{sim}_{\cos}(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|_2 \|\mathbf{v}\|_2} = \frac{\sum_{j=1}^{m} u_j v_j}{\sqrt{\sum_{j=1}^{m} u_j^2}\sqrt{\sum_{j=1}^{m} v_j^2}}.$$

The numerator is the *dot product*: multiply corresponding coordinates, then add the results. The denominator is the product of the two L_2 norms.

Example 32: Computing the similarity step by step

Take the two vectors from the right-hand graph, in the coordinate order (`three`, `we`):

$$\mathbf{u}_1 = (1, 1), \qquad \mathbf{u}_2 = (2, 1).$$

Step 1: compute the dot product. Multiply the counts for the same word and add:

$$\mathbf{u}_1 \cdot \mathbf{u}_2 = \underbrace{1 \cdot 2}_{\texttt{three}} + \underbrace{1 \cdot 1}_{\texttt{we}} = 3.$$

Each term compares the count of one word in the first document with the count of that same word in the second.

Step 2: compute the two L_2 norms.

$$\|\mathbf{u}_1\|_2 = \sqrt{1^2 + 1^2} = \sqrt{2}, \qquad \|\mathbf{u}_2\|_2 = \sqrt{2^2 + 1^2} = \sqrt{5}.$$

Step 3: divide the dot product by the product of the norms.

$$\mathrm{sim}_{\cos}(\mathbf{u}_1, \mathbf{u}_2) = \frac{3}{\sqrt{2}\sqrt{5}} = \frac{3}{\sqrt{10}} \approx 0.949.$$

The result is close to 1: the vectors point in similar directions, although their word proportions differ. This is the cosine of the angle of approximately 18.4° shown in the graph.

9.5.2 A limitation: matching words is not understanding meaning

Two texts can discuss the same subject while using different terms, such as `cat` and `feline`. In our representation these are separate coordinates: a shared meaning does not create a shared term. Conversely, the same word can have different meanings in different contexts. The count-based model does not resolve these ambiguities by itself. The next weighting step changes the importance of terms, but does not make distinct synonyms the same coordinate.

9.6 Why some words should count more than others

We can now compare documents using cosine similarity. However, in the count representation every occurrence has the same weight, regardless of which word it belongs to. A common word can therefore create a similarity that tells us little about the documents' topics.

Example 33: A similarity caused by a common word

Consider two short documents:

$$D_A: \quad \text{“the cat”}, \qquad D_B: \quad \text{“the dog”}.$$

Using the vocabulary order (`the`, `cat`, `dog`), we obtain

$$\mathbf{d}_A = (1, 1, 0), \qquad \mathbf{d}_B = (1, 0, 1).$$

Their dot product is

$$\mathbf{d}_A \cdot \mathbf{d}_B = \underbrace{1 \cdot 1}_{\texttt{the}} + \underbrace{1 \cdot 0}_{\texttt{cat}} + \underbrace{0 \cdot 1}_{\texttt{dog}} = 1.$$

Both vectors have L_2 norm $\sqrt{2}$, so

$$\mathrm{sim}_{\cos}(\mathbf{d}_A, \mathbf{d}_B) = \frac{1}{\sqrt{2}\sqrt{2}} = \frac{1}{2}.$$

The similarity comes entirely from `the`, a word that provides little information about what these documents discuss.

A first solution is to remove *stopwords*: very common words, such as articles and prepositions, that are often uninformative for the comparison. In the example, removing `the` leaves

$$D_A: \quad \text{“cat”}, \qquad D_B: \quad \text{“dog”}.$$

With the vocabulary order (`cat`, `dog`), the vectors become

$$\mathbf{d}_A = (1, 0), \qquad \mathbf{d}_B = (0, 1).$$

Their dot product is zero and both norms are 1, so their cosine similarity is now 0: they share no remaining terms.

This suggests a more general idea: words are not all equally informative. Instead of only removing certain words, we can assign different weights to the terms, so that informative words contribute more to the comparison.

9.7 Weighting terms by their rarity: IDF

A word that appears in almost every document does little to distinguish documents from one another. A word found in only a few documents can be more characteristic. For example, in a corpus of cooking texts, `food` might occur almost everywhere, whereas `guanciale` might appear only in recipes using that ingredient.

Example 34: Counting documents rather than occurrences

Suppose our corpus contains 100 documents:

Term	Documents containing the term
`food`	100
`pasta`	20
`guanciale`	2

These are counts of *documents*, not occurrences. If `guanciale` occurs ten times in one recipe, that recipe still contributes only 1 to the count of documents containing the term.

Step 1: count the documents containing each term. Let N be the total number of documents. For a term t, write $\mathrm{df}(t)$ for its *document frequency*: the number of documents containing it.

Step 2: compare the total with that count. Consider the ratio $N/\mathrm{df}(t)$. In our example,

$$\texttt{food}: \ \frac{100}{100} = 1, \qquad \texttt{pasta}: \ \frac{100}{20} = 5, \qquad \texttt{guanciale}: \ \frac{100}{2} = 50.$$

The rarer the term in the corpus, the larger this ratio.

Step 3: take the logarithm of the ratio. The logarithm reduces the differences between the weights while preserving their order: rarer terms still receive larger weights.

Definition 9.6: Inverse document frequency

For a term t appearing in at least one document of a corpus of N documents, its *inverse document frequency* is

$$\mathrm{IDF}(t) = \log\left(\frac{N}{\mathrm{df}(t)}\right),$$

where $\mathrm{df}(t)$ is the number of documents containing t.

Example 35: IDF weights for the cooking corpus

Using base-10 logarithms, the weights are

Term	$N/\mathrm{df}(t)$	$\mathrm{IDF}(t)$
`food`	$100/100 = 1$	$\log_{10}(1) = 0$
`pasta`	$100/20 = 5$	$\log_{10}(5) \approx 0.699$
`guanciale`	$100/2 = 50$	$\log_{10}(50) \approx 1.699$

Thus `guanciale` receives more weight than `pasta`. The word `food`, present in every document, receives weight 0: under this formula it provides no information for distinguishing the documents in this corpus.

9.8 Combining term frequency and rarity: TF-IDF

We now combine two pieces of information. The *term frequency* (TF) describes how often a term occurs in one particular document; IDF describes how rare that term is across the entire corpus. Here we use the raw occurrence count as TF, consistently with our count vectors.

Definition 9.7: TF-IDF weight

Let $\mathrm{TF}(t, D)$ be the number of occurrences of term t in document D. Its *TF-IDF weight* is

$$\text{TF-IDF}(t, D) = \mathrm{TF}(t, D) \cdot \mathrm{IDF}(t).$$

Example 36: From counts to TF-IDF weights

Step 1: keep the same corpus of 100 documents. Using the base-10 logarithms from the previous example,

$$\mathrm{IDF}(\texttt{food}) = 0, \qquad \mathrm{IDF}(\texttt{pasta}) \approx 0.699, \qquad \mathrm{IDF}(\texttt{guanciale}) \approx 1.699.$$

Step 2: choose a document D in this corpus. Suppose its counts for these terms are

Term	TF in document D
`food`	4
`pasta`	3
`guanciale`	2

In the coordinate order $(\texttt{food}, \texttt{pasta}, \texttt{guanciale})$, the corresponding count vector is $\mathbf{d} = (4, 3, 2)$. These are the three coordinates we are following in the example.

Step 3: multiply each count by its term's IDF.

Term	TF	IDF	TF-IDF
`food`	4	0	$4 \cdot 0 = 0$
`pasta`	3	≈ 0.699	$3 \log_{10}(5) \approx 2.097$
`guanciale`	2	≈ 1.699	$2 \log_{10}(50) \approx 3.398$

The weighted vector is therefore

$$\mathbf{d}_{\text{TF-IDF}} \approx (0,\ 2.097,\ 3.398).$$

Although `guanciale` occurs less often than `pasta` in this document, it receives a larger weight because it is rarer in the corpus. The term `food` receives weight zero because it appears in every document.

TF depends on the particular document being represented. IDF depends on the corpus: for a fixed corpus, a term has the same IDF in every document. We can then compare the weighted document vectors using the same cosine-similarity formula as before, provided neither weighted vector is zero.

9.9 Using document vectors for search

We can use TF-IDF vectors to build a search engine. A *query* is the text entered by the user. Given a query and a positive integer k, our goal is to return the k documents most similar to that query (or all documents if the corpus contains fewer than k). For example, $k = 3$ asks for the three highest-scoring results.

The first step is to represent the query as a vector in the *same space* as the documents: the same vocabulary, the same coordinate order and the same IDF weights.

Example 37: Turning a query into a vector

Suppose the user searches for

$$Q: \quad \text{“pasta guanciale”}.$$

Keep the coordinate order (`food`, `pasta`, `guanciale`) from the cooking example.

Step 1: count the query terms. The query contains no occurrence of `food`, one occurrence of `pasta` and one of `guanciale`. Its count vector is

$$\mathbf{q}_{\text{counts}} = (0, 1, 1).$$

Step 2: apply the same IDF weights as for the documents. Use the IDF values already computed from the corpus, rather than recomputing IDF from the query:

$$\mathbf{q}_{\text{TF-IDF}} \approx (0 \cdot 0,\ 1 \cdot 0.699,\ 1 \cdot 1.699) = (0,\ 0.699,\ 1.699).$$

As before, we display the three coordinates being followed in this example; any other vocabulary terms absent from the query have coordinate zero.

Step 3: compare the query with each document. For each document D_i, assign the score

$$\text{score}(D_i) = \text{sim}_{\cos}\big(\mathbf{q}_{\text{TF-IDF}}, \mathbf{d}_{i,\text{TF-IDF}}\big).$$

Here both vectors use the full common vocabulary and are assumed to be nonzero. A higher score means that the document's direction is closer to the query's direction. Sort the documents by decreasing score and return the first k.

The query is therefore treated as a small document. Representing it with the same coordinates and corpus-based IDF weights lets us use the same cosine-similarity calculation for retrieval.

9.9.1 The cost of searching every document

The simplest search procedure follows four steps:

1. Compute cosine similarity between the query and every document.
2. Store each document's score.

3. Sort the documents by decreasing score.
4. Return the first k results.

Suppose the corpus contains N documents and the vocabulary contains m terms. We analyse the direct implementation that scans all m coordinates of each document vector.

First cost: computing all scores. Each vector has m coordinates. Scanning all of them to compare the query with one document takes $\Theta(m)$ time. Repeating this for all N documents takes

$$\Theta(Nm).$$

Example 38: A large corpus

With one million documents and a vocabulary of one hundred thousand terms, the direct implementation examines

$$10^6 \cdot 10^5 = 10^{11}$$

pairs of coordinates for every query. Each pair contains the query weight and the document weight for the same vocabulary term.

Second cost: sorting the scores. One simple approach repeatedly finds the largest remaining score, places its document next in the ordering and removes it from consideration. Producing the complete ordering this way scans

$$N + (N-1) + \cdots + 1 = \frac{N(N+1)}{2}$$

scores, so it takes $\Theta(N^2)$ time.

An efficient comparison-based sorting algorithm sorts the N scores in $O(N \log N)$ time. Thus computing all scores and sorting them gives the overall bound

$$O(Nm + N \log N).$$

This procedure works, but repeats substantial work whenever a new query arrives. The next question is which work can be performed once, while preparing the documents, and which must wait until the query is known.

9.9.2 Preparing the corpus offline, answering queries online

The corpus is usually available before the user submits a query. We can therefore prepare information about the documents in advance, instead of repeating all the preparation for every search.

Definition 9.8: Offline and online phases

The *offline phase* processes the corpus before a particular query is known, storing information useful for future searches. The *online phase* starts when a query arrives and uses that stored information to compute the results.

Here "offline" means independent of the current query, rather than disconnected from the Internet. For our vector-based search, we can prepare the vocabulary, document counts, IDF weights and document TF-IDF vectors offline. We can also store their L_2 norms, or their normalized vectors. Online, we process the query, construct its vector using the stored vocabulary and IDF, and compute the results.

Example 39: Many users searching the same corpus

Suppose one user searches for "pasta guanciale" and another for "tomato restaurant". Their query vectors differ, but the document vectors and corpus-based IDF weights are the same. We prepare those once and reuse them for both searches. If the corpus changes, the stored information must be updated as appropriate.

Precomputing document vectors saves preparation work, but scanning every document still has a cost. To understand how a different data organization can help, we now consider a Boolean version of search.

9.10 Boolean AND search

Suppose the query is "pasta guanciale". This time we want every document containing *both* words. A document containing only `pasta` does not qualify, however often that word occurs.

Definition 9.9: Boolean AND query

Represent each document D_i by its set of terms T_i, and let Q be a nonempty set of query terms. A Boolean AND query returns

$$R(Q) = \{i : Q \subseteq T_i\}.$$

Thus a document qualifies precisely when it contains every query term.

This search asks a membership question, rather than assigning cosine scores and selecting the top k. Its output is the set of all qualifying document identifiers.

9.10.1 A direct scan

A first approach to answering this query is to visit each document, check whether it contains every query term, and add its identifier to the result if it does. For a large corpus this inspects many documents that contain none of the query terms. We would like to find the relevant documents without starting a new scan of the entire corpus each time.

9.11 The inverted index

The usual document representation answers: "Which terms occur in this document?" We now reverse the direction and store the answer to: "Which documents contain this term?"

Definition 9.10: Inverted index and posting list

An *inverted index* maps each term t to its *posting list* $L(t)$: the sorted list of identifiers of documents containing t. Each document identifier appears at most once in a term's list, regardless of how often that term occurs in the document.

Example 40: Building the index from six documents

Use letters to represent terms, and number the documents from 1 to 6. The original representation is

Document identifier	Terms in the document
1	A, B, C, D
2	A, B, E, G
3	B, D, F
4	A, C, D, E
5	A, C, G
6	C, D, G

Step 1: collect the documents containing A. Reading the table, A occurs in documents $1, 2, 4, 5$, so $L(A) = [1, 2, 4, 5]$.

Step 2: repeat for every term. The complete inverted index for these six documents is

Term	Posting list
A	$[1, 2, 4, 5]$
B	$[1, 2, 3]$
C	$[1, 4, 5, 6]$
D	$[1, 3, 4, 6]$
E	$[2, 4]$
F	$[3]$
G	$[2, 5, 6]$

For a single-term query E, the answer is immediately $\{2, 4\}$. We look up $L(E)$ rather than checking all six documents.

We build the inverted index offline. One way is to process documents in increasing identifier order and append each identifier to the lists of its distinct terms; the lists are then already sorted. Online, a query only needs the lists for its own terms.

9.11.1 AND means intersecting posting lists

For the query $Q = \{A, C, E\}$, a document must occur in all three lists:

$$R(Q) = L(A) \cap L(C) \cap L(E).$$

Here we use intersection to mean the identifiers shared by the lists. Using Example 40,

$$\{1, 2, 4, 5\} \cap \{1, 4, 5, 6\} = \{1, 4, 5\}, \qquad \{1, 4, 5\} \cap \{2, 4\} = \{4\}.$$

The answer is document 4. Looking back at its row confirms that it contains A, C, E. If any query term has no posting list, no document contains that term and the AND result is empty.

9.12 Intersecting sorted lists with pointers

How can we find the intersection without repeatedly searching one list for every element of another? Sorting lets us move through the lists from left to right, using a *pointer* to mark the current position in each list.

Example 41: Following the pointers

The query is $Q = \{A, C, E\}$: we want documents containing all three terms. The posting lists are

$$L(A) = [1, 2, 4, 5], \quad L(C) = [1, 4, 5, 6], \quad L(E) = [2, 4].$$

Each number identifies a document. Put a pointer at the first element of each list; the boxed entries below show the current positions.

Step 1: the pointers read $1, 1, 2$.

$$L(A) = [\boxed{1}, 2, 4, 5],$$
$$L(C) = [\boxed{1}, 4, 5, 6],$$
$$L(E) = [\boxed{2}, 4].$$

Document 1 contains A and C, but not E. In $L(E)$ we are already at 2; since the list is sorted, no later entry can be 1. Thus 1 cannot be a result. Advance the two pointers reading 1 (those in $L(A)$ and $L(C)$), leaving the pointer in $L(E)$ at 2.

Step 2: the pointers now read $2, 4, 2$.

$$L(A) = [1, \boxed{2}, 4, 5],$$
$$L(C) = [1, \boxed{4}, 5, 6],$$
$$L(E) = [\boxed{2}, 4].$$

Document 2 contains A and E, but not C. In $L(C)$ we are already at 4, so we cannot find 2 further along that list. Discard 2: advance the pointers in $L(A)$ and $L(E)$, leaving the pointer in $L(C)$ at 4.

Step 3: the pointers now read $4, 4, 4$.

$$L(A) = [1, 2, \boxed{4}, 5],$$
$$L(C) = [1, \boxed{4}, 5, 6],$$
$$L(E) = [2, \boxed{4}].$$

All three pointers read the same identifier: document 4 contains A, C and E. Add it to the result, giving $R = [4]$, then advance all three pointers.

Step 4: the list for E is exhausted.

$$L(A) = [1, 2, 4, \boxed{5}],$$
$$L(C) = [1, 4, \boxed{5}, 6],$$
$$L(E) = [2, 4] \quad \text{(pointer past the last entry)}.$$

The other pointers read 5, but $L(E)$ has no entries left. No further identifier can occur in all three lists, so we stop. The final result is $R = [4]$: return document D_4.

Summary of the pointer movements.

Step	$L(A)$	$L(C)$	$L(E)$	Action
1	1	1	2	Advance the pointers at 1.
2	2	4	2	Advance the pointers at 2.
3	4	4	4	Output 4; advance all three.
4	5	5	exhausted	Stop.

For a fixed small number of lists, this gives a simple rule: if all current identifiers agree, output that identifier and advance every pointer; otherwise advance the pointers at the smallest current identifier. No pointer needs to move backwards.

9.12.1 A two-list intersection algorithm

For two lists, there are only three cases: the current identifiers are equal, the left one is smaller, or the right one is smaller. The following algorithm makes each case explicit.

Algorithm 9.1: Intersect two sorted posting lists

```
1: function INTERSECT(L, M)
2:     i ← 1, j ← 1, R ← [ ]
3:     while i ≤ |L| and j ≤ |M| do
4:         if L[i] = M[j] then
5:             Append L[i] to R
6:             i ← i + 1, j ← j + 1
7:         else if L[i] < M[j] then
8:             i ← i + 1
9:         else
10:            j ← j + 1
11:        end if
12:    end while
13:    return R
14: end function
```

Every iteration advances at least one pointer, and each pointer crosses its list at most once. Thus intersecting lists of lengths a and b takes $O(a+b)$ time, with $\Theta(a+b)$ in the worst case. It can finish sooner if one list is exhausted early.

9.12.2 Query cost: the lengths of the relevant lists

We now measure the work needed to answer a query after the inverted index has already been built. Let $n_t = |L(t)|$ be the length of the posting list for query term t: the number of documents containing that term. We work with the lists selected by the query.

For several terms, we can apply the two-list algorithm repeatedly. Start with the shortest posting list and intersect the current result with each of the other lists. Stop if the result becomes empty.

Example 42: The cost of a three-term query

Take $Q = \{A, C, E\}$, with

$$L(A) = [1, 2, 4, 5], \quad L(C) = [1, 4, 5, 6], \quad L(E) = [2, 4].$$

Step 1: count the lengths of the relevant lists.

Term	Posting-list length
A	$n_A = 4$
C	$n_C = 4$
E	$n_E = 2$

Their sum is $n_A + n_C + n_E = 4 + 4 + 2 = 10$. The lists for B, D, F, G are not needed for this query.

Step 2: start with the shortest list, $L(E)$, and intersect it with $L(A)$.

$$[2, 4] \cap [1, 2, 4, 5] = [2, 4].$$

The pairs of current identifiers compared by the two pointers are

$$(2,1) \longrightarrow (2,2) \longrightarrow (4,4).$$

First advance past 1 in $L(A)$. Then output 2 and advance both pointers; next output 4 and advance both again. The first list is now exhausted. We made three comparisons and obtained $R = [2,4]$.

Step 3: intersect the intermediate result with $L(C)$.

$$[2,4] \cap [1,4,5,6] = [4].$$

This time the comparisons are

$$(2,1) \longrightarrow (2,4) \longrightarrow (4,4).$$

Advance past 1 in $L(C)$, then past 2 in the intermediate result. Finally output 4 and advance both pointers. The intermediate list is exhausted. These are three more comparisons, and the final answer is document D_4.

Step 4: relate the work to the list lengths. The first intersection has the bound $O(2+4)$; the second also has the bound $O(2+4)$. The intermediate list never grows: it starts with at most two identifiers and intersections can only remove them. We performed six identifier comparisons in this example.

The general bound below is proportional to the sum of the original posting-list lengths, which here is 10.

Why starting with the shortest list gives the total bound. The intermediate result is always a subset of the shortest initial list, so its length never exceeds that list's length.

Let s be the length of the shortest initial list and n_t the length of the next posting list. The intermediate result has length at most s, so intersecting it with that next list costs $O(s+n_t)$. Since the next list cannot be shorter than the shortest initial list, $s \leq n_t$. Therefore

$$s + n_t \leq 2n_t, \qquad O(s+n_t) = O(n_t).$$

To see the total bound, let $r = |Q|$, the number of query terms and hence the number of posting lists. One list is used as the starting point, leaving $r - 1$ lists to intersect with the current result. Since each of these other lists has length at least s,

$$(r-1)s \leq \sum_{t \in Q \setminus \{t_*\}} n_t,$$

where t_* is the term with the shortest list. Consequently the total intersection time is

$$O\left(\sum_{t \in Q} n_t\right).$$

To retrieve the lists, look up each of the r query terms in the inverted index. With a hash-table dictionary, each lookup takes $O(1)$ expected time, giving $O(r)$ expected time for all lookups. Retrieving a list here means obtaining a reference to the stored list, rather than copying or scanning its entries. Scanning the entries is counted in the intersection cost. The overall expected query cost is therefore

$$O\left(\underbrace{r}_{\text{retrieve the lists}} + \underbrace{\sum_{t \in Q} n_t}_{\text{intersect the lists}}\right).$$

In a much larger corpus, rare query terms can have short lists, making the online search far cheaper than a full corpus scan. If the terms occur in almost every document, their lists are long and the advantage is smaller.

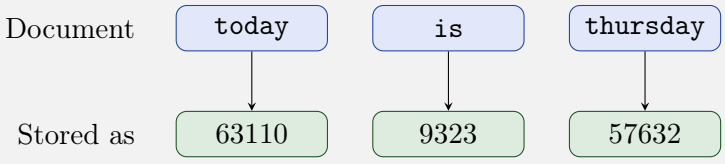
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9.13 Terms as integer identifiers

Comparing numbers is simpler and cheaper than comparing words. Offline, we can sort the vocabulary and give each term an integer *identifier*, its position in that order. A document is then stored as a sequence of identifiers, and the inverted index uses these numbers as keys.

Example 43: From words to identifiers

In a sorted English vocabulary, `a` receives 0, `aa` receives 1, and so on. Suppose `is`, `thursday` and `today` receive the identifiers below.



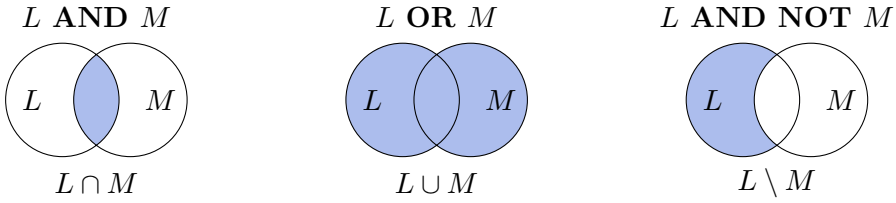
In the examples we keep letters such as A, B, C for the terms, which are easier to read than five-digit numbers.

9.14 Boolean queries with OR and NOT

An AND query intersects posting lists. A Boolean query can also use OR and NOT. For example, someone looking for fettuccine with veal, but not fettuccine Alfredo, could search for

`fettuccine` AND `vitello` AND NOT `alfredo`.

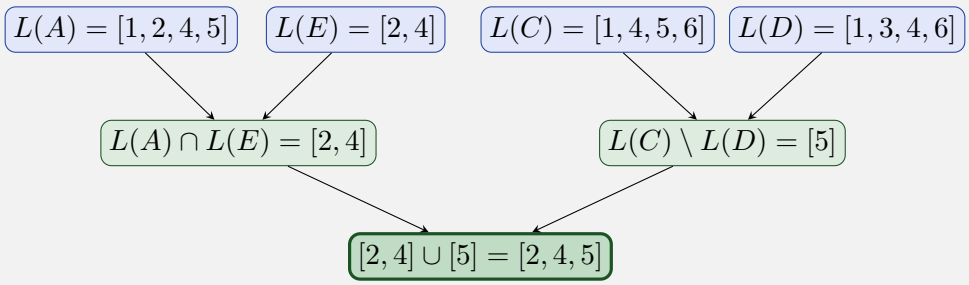
For two posting lists L and M, each operation corresponds to a set operation on document identifiers:



A query containing only NOT, such as NOT D, would return almost the whole corpus. NOT is therefore used together with AND, as in A AND NOT D.

Example 44: Evaluating (A AND E) OR (C AND NOT D)

Using the inverted index of Example 40, evaluate the two parts in parentheses first, then combine them with OR.



9.14.1 The same pointers, different rules

All three operations scan the two sorted lists from left to right with one pointer each, as in Algorithm 9.1. Only the action taken at each step changes:

Query	Same identifier	Different identifiers	One list ends
L AND M	output it; advance both	advance the smaller	stop
L OR M	output it once; advance both	output the smaller; advance it	copy the rest of the other list
L AND NOT M	skip it; advance both	output the smaller only if it is in L; advance it	if M ends, copy the rest of L; if L ends, stop

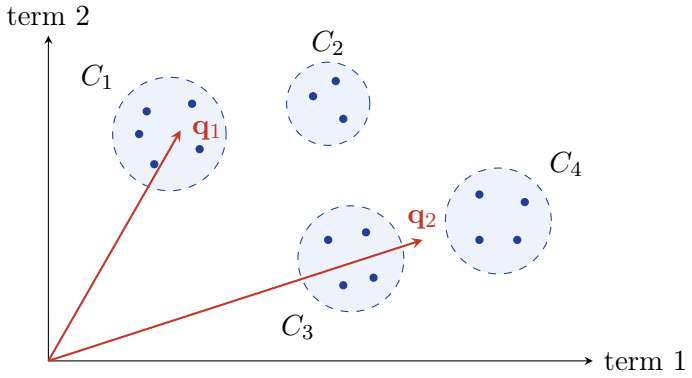
Every pointer still moves only forwards, so each operation costs $O(|L| + |M|)$. A query combining several operations therefore costs $O\left(\sum_{t \in Q} n_t\right)$, as for AND.

9.15 Faster ranked search

We return to ranked search (Section 9.9): given a query vector $\mathbf{q}$ and an integer K, return the K documents with the largest cosine similarity to $\mathbf{q}$. Scanning every document costs $\Theta(Nm)$. Can we avoid looking at every document and every term?

9.15.1 Searching inside one cluster

Offline, we can group the documents into *clusters* of similar documents and give each cluster a representative vector. Online, we first find the cluster whose representative is closest to the query, then compute cosine similarities only for the documents in that cluster.



For $\mathbf{q}_1$, the closest documents are clearly in C_1. The query $\mathbf{q}_2$ lies between C_3 and C_4, so its best documents may belong to either cluster: searching only one of them can miss good results. Clustering makes search faster, but the answer is only approximate.

9.15.2 Scoring only documents that share a query term

Write $d_{i,t}$ for the weight of term t in document D_i and q_t for its weight in the query. All weights are nonnegative, so the dot product

$$\mathbf{q} \cdot \mathbf{d}_i = \sum_{t \in Q} q_t \, d_{i,t}$$

is positive only if D_i contains at least one query term. Every other document has similarity 0. The documents that can score are exactly those in the posting lists of the query terms.

To compute the weights from the index, we extend each posting list: each entry stores a pair $(i, \mathrm{TF}(t, D_i))$ instead of the identifier alone. The cosine similarity then splits into a part computed online and parts that do not depend on the documents or on the query:

$$\mathrm{sim}_{\cos}(\mathbf{q}, \mathbf{d}_i) = \frac{\overbrace{\sum_{t \in Q} q_t \, d_{i,t}}^{\text{online, from the posting lists}}}{\underbrace{\|\mathbf{q}\|_2}_{\text{same for every } i} \quad \underbrace{\|\mathbf{d}_i\|_2}_{\text{stored offline}}}.$$

- $\|\mathbf{q}\|_2$ is the same for every document, so it does not change the ranking and we can ignore it.
- $\|\mathbf{d}_i\|_2$ does not depend on the query: we compute it offline and store it with the document.
- $\mathrm{IDF}(t) = \log(N/n_t)$ depends only on the length n_t of the posting list $L(t)$, so it is also available offline.

Algorithm 9.2: Top-K search with an inverted index

```
function TopK(Q, K)
    S ← an empty map from documents to scores
    for each term t ∈ Q do
        for each pair (i, tf) in L(t) do
            S[i] ← S[i] + q_t · tf · IDF(t)        ▷ missing entries start at 0
        end for
    end for
    for each document i in S do
        S[i] ← S[i] / ||d_i||_2                    ▷ norm stored offline
    end for
    return the K documents with the largest scores in S
end function
```

The loops read each entry of the query's posting lists once, in $O\big(\sum_{t\in Q} n_t\big)$ time. Only the $c \le \sum_{t\in Q} n_t$ documents found in those lists receive a score, and sorting them costs $O(c \log c)$. When the query terms are rare, this is far less than the $\Theta(Nm + N \log N)$ needed to score and sort every document.

Example 45: Scoring the query $Q = \{C, E\}$

For simplicity, give every term the same weight: $q_t = 1$ for the query terms and $d_{i,t} = \mathrm{TF}(t, D_i)$. Take the six documents of Example 40, now with repeated terms:

$$D_1 : A\,B\,C\,D, \qquad D_2 : A\,B\,E\,G, \qquad D_3 : B\,D\,F,$$
$$D_4 : A\,C\,C\,D\,D\,E, \qquad D_5 : A\,C\,G, \qquad D_6 : C\,C\,C\,D\,G.$$

The posting lists of the query terms store (document, TF) pairs:

$$L(C) = [(1,1), (4,2), (5,1), (6,3)], \qquad L(E) = [(2,1), (4,1)].$$

Add up the TF values of each document found in these two lists, then divide by its stored norm:

Document	$\mathrm{TF}(C) + \mathrm{TF}(E)$	$\|\mathbf{d}_i\|_2$ (offline)	Score
1	$1 + 0 = 1$	$\sqrt{4} = 2$	0.50
2	$0 + 1 = 1$	$\sqrt{4} = 2$	0.50
4	$2 + 1 = 3$	$\sqrt{10} \approx 3.16$	**0.95**
5	$1 + 0 = 1$	$\sqrt{3} \approx 1.73$	0.58
6	$3 + 0 = 3$	$\sqrt{11} \approx 3.32$	**0.90**

For $K = 2$ we return D_4 and D_6. Document 3 is never examined, because it contains neither C nor E. Dividing every score by $\|\mathbf{q}\|_2 = \sqrt{2}$ would give the exact cosines, with the same ranking.

With TF-IDF weights, the procedure is the same: multiply each TF by IDF(t), computed from the list length n_t, and store the norms of the TF-IDF vectors. In the example, IDF(E) = log(6/2) is larger than IDF(C) = log(6/4), so E would count more than C.

10 Sorting

Start of lecture 26 October

10.1 The sorting problem

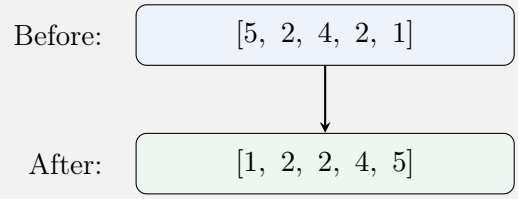
Sorting means arranging elements in order. We begin with a simple goal: given a list of numbers, arrange them from smallest to largest.

Problem: Sorting

Input: an array A of numbers, not necessarily sorted.

Output: an array containing the same elements as A in nondecreasing order, preserving every occurrence of each value.

Example 46: The same numbers, in a different order



The result must satisfy two conditions:

1. Each number is less than or equal to the next one.
2. The elements remain the same: only their order changes. In particular, both occurrences of 2 must remain in the result.

With five numbers, we can easily do this by hand. A computer needs precise instructions that also work for millions of elements. We will study different algorithms for the same sorting problem, examining how they work and how much time they require.

10.2 MergeSort

The first sorting algorithm we study is *MergeSort*. For example, we may start from

$$A = [2, 10, 7, 16, 3, 8, 9, 1, 11, 13, 14, 20, 17].$$

10.2.1 How Merge works: always take the smaller current element

Before studying the complete algorithm, we focus our attention on the function MERGE, a building block used by MergeSort.

Problem: Merging two sorted arrays

Input: two arrays A_1 and A_2, each already sorted in nondecreasing order.

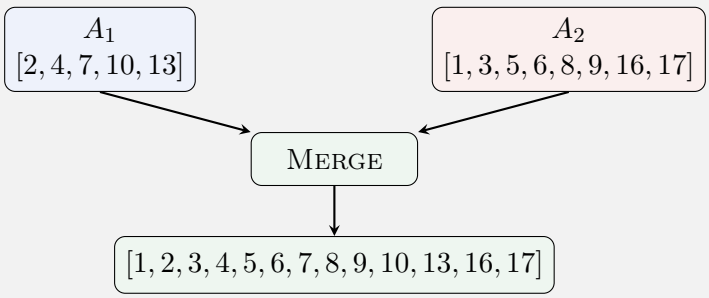
Output: one sorted array containing all the elements of both arrays, preserving every occurrence of each value.

Example 47: The input and output of Merge

Consider the following separate example of a call to MERGE(A_1, A_2):

$$A_1 = [2, 4, 7, 10, 13], \qquad A_2 = [1, 3, 5, 6, 8, 9, 16, 17].$$

We want one array containing all their values, in order:



The first array has 5 elements and the second has 8, so the output must contain $5 + 8 = 13$ elements.

Because both input arrays are sorted, the smallest element still available in each array is its first unprocessed element. To choose the next output value, we only need to compare those two elements.

Place a pointer i at the start of A_1 and a pointer j at the start of A_2. Create an empty output array B. At each step:

1. Compare $A_1[i]$ and $A_2[j]$.
2. Append the smaller value to B.
3. Advance only the pointer of the array from which we took it.

If the two values are equal, take the one from A_1 first. The other occurrence remains available in A_2, so neither occurrence is lost.

Example 48: Following Merge on the two arrays

Start with

$$A_1 = [2, 4, 7, 10, 13], \qquad A_2 = [1, 3, 5, 6, 8, 9, 16, 17], \qquad B = [\,].$$

Step 1: compare 2 and 1.

$$A_1 = [\boxed{2}, 4, 7, 10, 13], \qquad A_2 = [\boxed{1}, 3, 5, 6, 8, 9, 16, 17].$$

Take 1 from A_2: $B = [1]$. Advance its pointer to 3; the pointer in A_1 stays at 2.

Step 2: compare 2 and 3.

$$A_1 = [\boxed{2}, 4, 7, 10, 13], \qquad A_2 = [1, \boxed{3}, 5, 6, 8, 9, 16, 17].$$

Take 2 from A_1: $B = [1, 2]$. Advance its pointer to 4.

Step 3: compare 4 and 3.

$$A_1 = [2, \boxed{4}, 7, 10, 13], \qquad A_2 = [1, \boxed{3}, 5, 6, 8, 9, 16, 17].$$

Take 3 from A_2: $B = [1, 2, 3]$. Advance its pointer to 5.

Continue with the same rule. The following rows show the current values before each remaining comparison and the output after taking the smaller one:

$A_1[i]$	$A_2[j]$	Take	Output B
4	5	4	$[1, 2, 3, 4]$
7	5	5	$[1, 2, 3, 4, 5]$
7	6	6	$[1, 2, 3, 4, 5, 6]$
7	8	7	$[1, 2, 3, 4, 5, 6, 7]$
10	8	8	$[1, 2, 3, 4, 5, 6, 7, 8]$
10	9	9	$[1, 2, 3, 4, 5, 6, 7, 8, 9]$
10	16	10	$[1, 2, 3, 4, 5, 6, 7, 8, 9, 10]$
13	16	13	$[1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 13]$

When one input is exhausted, copy the rest of the other. After taking 13, A_1 is exhausted. The remaining values in A_2 are 16, 17. They are already sorted and can be appended directly:

$$B = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 13, 16, 17].$$

The algorithm below uses indices starting at 1. The output array B has space for $|A_1|+|A_2|$ elements, and k indicates the next output position to fill.

Algorithm 10.1: Merge two sorted arrays

```
1: function MERGE(A_1, A_2)
2:     n_1 ← |A_1|, n_2 ← |A_2|
3:     Allocate an array B of length n_1 + n_2
4:     i ← 1, j ← 1, k ← 1
5:     while i ≤ n_1 and j ≤ n_2 do
6:         if A_1[i] ≤ A_2[j] then
7:             B[k] ← A_1[i]
8:             i ← i + 1
9:         else
10:            B[k] ← A_2[j]
11:            j ← j + 1
12:        end if
13:        k ← k + 1
14:    end while
15:    while i ≤ n_1 do
16:        B[k] ← A_1[i]
17:        i ← i + 1, k ← k + 1
18:    end while
19:    while j ≤ n_2 do
20:        B[k] ← A_2[j]
21:        j ← j + 1, k ← k + 1
22:    end while
23:    return B
24: end function
```

10.2.2 Why the result is sorted, and how much Merge costs

At every step, the next value must be the smaller of the two current elements: all later elements of each input are at least as large as that input's current element. Appending the smaller one therefore keeps B sorted.

Let $n_1 = |A_1|$ be the number of elements in the first array and $n_2 = |A_2|$ the number of elements in the second. Merge copies each of these elements into B exactly once: n_1 copies come from A_1 and n_2 from A_2, for a total of $n_1 + n_2$ copies. The pointers only move forwards, and each copy and pointer update takes constant time. Its running time is therefore

$$\boxed{T_{\text{Merge}}(n_1, n_2) = \Theta(n_1 + n_2).}$$

The separate output array uses $\Theta(n_1 + n_2)$ additional space; the three indices require only constant extra space.

10.3 MergeSort: split, sort the halves, and merge

Merge combines two arrays that are already sorted. To sort an arbitrary array A, we first split it into two halves and sort each half using MergeSort itself. We then pass the two sorted results to Merge.

When can we stop splitting? An array with zero or one element is already sorted, so we return it immediately. This is the base case.

For $n > 1$, let $m = \lceil n/2 \rceil$. Using indices starting at 1, the two halves are

$$A_1 = A[1, \ldots, m], \qquad A_2 = A[m+1, \ldots, n].$$

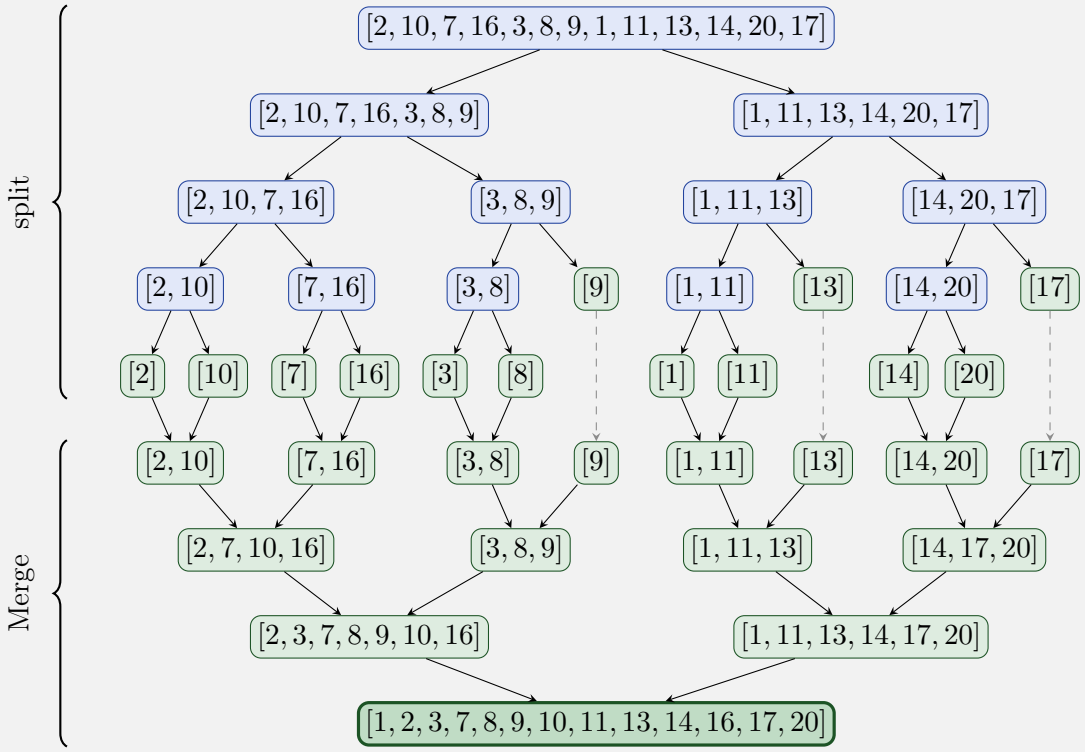
The first half contains m elements and the second contains $n - m$. For our 13-element input, $m = 7$: the halves contain positions 1 through 7 and positions 8 through 13.

Algorithm 10.2: MergeSort

```
1:  function MergeSort(A)
2:      n ← |A|                                   ▷ Θ(1)
3:      if n ≤ 1 then                             ▷ Θ(1)
4:          return A                              ▷ Θ(1)
5:      end if
6:      m ← ⌈n/2⌉                                 ▷ Θ(1)
7:      A1 ← A[1, ..., m]                         ▷ Θ(m) to copy
8:      A2 ← A[m + 1, ..., n]                     ▷ Θ(n − m) to copy
9:      A1 ← MergeSort(A1)                        ▷ T(m)
10:     A2 ← MergeSort(A2)                        ▷ T(n − m)
11:     return Merge(A1, A2)                      ▷ Θ(n)
12: end function
```

Example 49: MergeSort on the 13-element array

MergeSort on $A = [2, 10, 7, 16, 3, 8, 9, 1, 11, 13, 14, 20, 17]$. Blue arrays are still unsorted; green arrays are sorted.



10.3.1 Running time: expanding the recurrence

Let $T(n)$ be the number of steps for an array of n elements. Assume first that n is a power of 2, so $m = n - m = n/2$. Counting the steps of Algorithm 10.2 gives a *recurrence*:

$$T(n) = \begin{cases} c_1, & n \leq 1, \\ 2T(n/2) + c_2 n + c_3, & n > 1, \end{cases}$$

Here c_1 is the base-case cost, $c_2 n$ accounts for copying the halves and merging, and c_3 is the remaining constant work. The constants are positive and independent of n.

We solve this recurrence by repeatedly substituting it into itself. As in the earlier recurrence expansions, a term and its replacement are underlined in the same colour.

First expansion. Apply the recurrence to an input of size $n/2$:

$$T(n/2) = 2T(n/4) + c_2\frac{n}{2} + c_3.$$

Substitute this expression for $T(n/2)$ and distribute the outer 2:

$$\begin{aligned} T(n) &= 2\,T(n/2) + c_2 n + c_3 \\ &= 2\left[2T(n/4) + c_2\frac{n}{2} + c_3\right] + c_2 n + c_3 \\ &= 4T(n/4) + c_2 n + 2c_3 + c_2 n + c_3 \\ &= 4T(n/4) + 2c_2 n + 3c_3. \end{aligned}$$

There are now four subproblems of size $n/4$. The two linear contributions add up to $2c_2 n$, and the constant contributions add up to $3c_3$.

Second expansion. Now replace $T(n/4)$ using the same recurrence:

$$\begin{aligned} T(n) &= 4\,T(n/4) + 2c_2 n + 3c_3 \\ &= 4\left[2T(n/8) + c_2\frac{n}{4} + c_3\right] + 2c_2 n + 3c_3 \\ &= 8T(n/8) + c_2 n + 4c_3 + 2c_2 n + 3c_3 \\ &= 8T(n/8) + 3c_2 n + 7c_3. \end{aligned}$$

Third expansion.

$$\begin{aligned} T(n) &= 8\,T(n/8) + 3c_2 n + 7c_3 \\ &= 8\left[2T(n/16) + c_2\frac{n}{8} + c_3\right] + 3c_2 n + 7c_3 \\ &= 16T(n/16) + 4c_2 n + 15c_3. \end{aligned}$$

The general pattern. Each substitution doubles the number of subproblems and halves their size. At each level, the linear work adds up to $c_2 n$:

$$1 \cdot c_2 n = 2 \cdot c_2\frac{n}{2} = 4 \cdot c_2\frac{n}{4} = \cdots = c_2 n.$$

After expanding i levels, we therefore have

$$\boxed{T(n) = 2^i T\left(\frac{n}{2^i}\right) + i\,c_2 n + (2^i - 1)c_3.}$$

Here $2^i T(n/2^i)$ is the work still to be expanded, $i\,c_2 n$ accounts for the i levels already expanded, and the constant work comes from $1 + 2 + 4 + \cdots + 2^{i-1} = 2^i - 1$ calls at those levels.

When do we stop? We reach the base case when each remaining subproblem has size 1:

$$\frac{n}{2^i} = 1 \quad \Longrightarrow \quad 2^i = n \quad \Longrightarrow \quad i = \log_2 n.$$

Substituting $i = \log_2 n$ and $T(1) = c_1$ gives

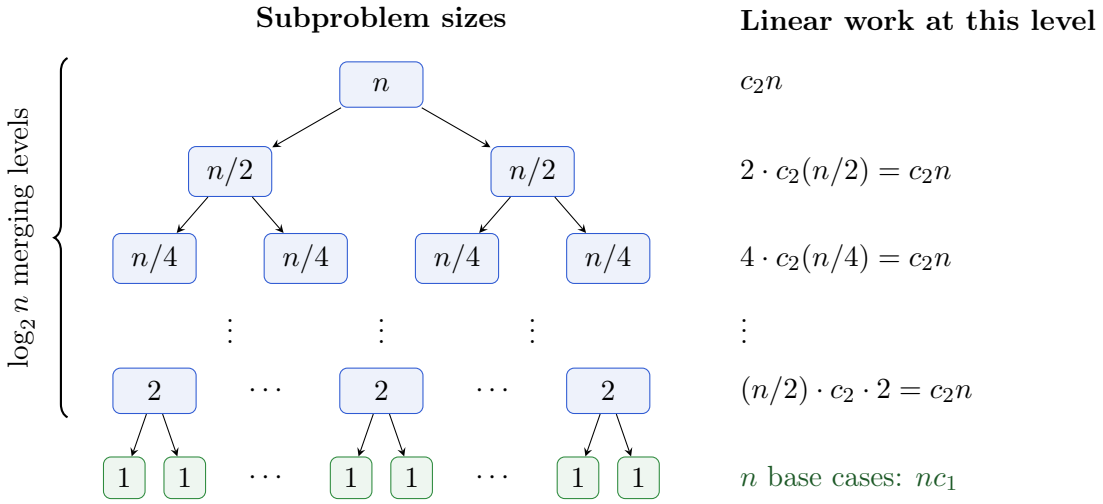
$$\begin{aligned} T(n) &= nT(1) + c_2 n \log_2 n + (n-1)c_3 \\ &= c_1 n + c_2 n \log_2 n + c_3(n-1). \end{aligned}$$

The first and last terms are linear in n. The dominant term is $c_2 n \log_2 n$, so

$$\boxed{T(n) = \Theta(n \log n).}$$

For a general n, the two recursive sizes are $\lceil n/2 \rceil$ and $\lfloor n/2 \rfloor$. The number of levels is $\lceil \log_2 n \rceil$, and the same asymptotic bound holds.

Visualizing the work at each level. Each box below represents a recursive call and shows the number of elements it sorts. The subarrays at each merging level therefore contain n elements altogether.



Each merging level contributes $c_2 n$ linear work. There are $\log_2 n$ such levels, giving $c_2 n \log_2 n$ altogether. At the bottom, the n single-element inputs contribute nc_1.

11 Randomized Algorithms

11.1 Making random choices inside an algorithm

A deterministic algorithm follows the same execution whenever it receives the same input. We now allow the algorithm to make random choices during its execution. For example, a sorting algorithm may choose a different element to work with first on each run.

Definition 11.1: Randomized algorithm

A *randomized algorithm* uses random choices as part of its computation. Even for a fixed input, different choices may lead to different executions and running times.

For randomized QuickSort, every execution will still return a sorted array. What changes with the random choices is the amount of work needed to obtain it. To describe this cost, we first need to understand an average taken over random outcomes.

11.2 Expected value: averaging over random outcomes

Imagine rolling a fair die many times and averaging the results. Each of the six faces appears with probability 1/6, so each face contributes equally to the average.

Definition 11.2: Random variable and expected value

A discrete *random variable* X assigns a numerical value to each random outcome. Its *expected value* is the weighted average of its possible values:

$$\mathbb{E}[X] = \sum_{x \in \text{range}(X)} x \, \mathbb{P}(X = x).$$

Each value x is weighted by its probability of occurring.

Example 50: One die, then two dice

Step 1: one fair die. Let Y be the number on its upper face. The possible values are $1, 2, 3, 4, 5, 6$, each with probability $1/6$. Thus

$$\begin{aligned}\mathbb{E}[Y] &= 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6} \\ &= \frac{1+2+3+4+5+6}{6} = \frac{21}{6} = 3.5.\end{aligned}$$

Step 2: the sum of two fair independent dice. Let $X = Y_1 + Y_2$. There are $6 \cdot 6 = 36$ equally likely ordered pairs of faces. The table shows the sum for each pair:

$Y_1 \backslash Y_2$	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

The sum 2 occurs only for $(1, 1)$, whereas 3 occurs for $(1, 2)$ and $(2, 1)$. The sum 7 occurs for six different pairs. Therefore

$$\mathbb{P}(X = 2) = \frac{1}{36}, \qquad \mathbb{P}(X = 3) = \frac{2}{36}, \qquad \mathbb{P}(X = 7) = \frac{6}{36}.$$

The possible sums are not equally likely. We can compute their weighted average using the definition. A shorter calculation uses *linearity of expectation*: the expected value of a sum is the sum of the expected values. Hence

$$\mathbb{E}[X] = \mathbb{E}[Y_1 + Y_2] = \mathbb{E}[Y_1] + \mathbb{E}[Y_2] = 3.5 + 3.5 = 7.$$

For a randomized algorithm, the running time on a fixed input is also a random variable. Its expected running time averages over the algorithm's random choices.

11.3 QuickSort: choose a pivot and partition

QuickSort sorts an array by choosing one of its elements as a *pivot*. It compares the remaining elements with that pivot and separates them into two smaller arrays.

Definition 11.3: Pivot and partition

Choose an index p uniformly from $\{1, \ldots, n\}$ and let $v = A[p]$ be the pivot value. Excluding $A[p]$, form

$$A_1 = [\text{elements of } A \text{ that are } \leq v], \qquad A_2 = [\text{elements of } A \text{ that are } > v].$$

Every remaining element goes into exactly one of these two arrays. This separation is called *partitioning*.

Now sort A_1 and A_2 recursively. If their sorted results are A_1' and A_2', the answer is their concatenation with the pivot between them:

$$[\, A_1',\ v,\ A_2' \,].$$

Every value in A_1' is at most v, and every value in A_2' is greater than v. Once the two parts are sorted, placing them in this order produces a sorted array. As before, an array with zero or one element is already sorted and is returned immediately.

11.3.1 The QuickSort algorithm

Indices start at 1. Appending to the temporary arrays is assumed to take amortized constant time. The final concatenation copies the two sorted parts and the pivot into the returned array.

Algorithm 11.1: Randomized QuickSort

```
1:  function QuickSort(A)
2:      n ← |A|                                              ▷ Θ(1)
3:      if n ≤ 1 then
4:          return A
5:      end if
6:      Choose p uniformly from {1, ..., n}
7:      v ← A[p]
8:      A_1 ← [ ], A_2 ← [ ]
9:      for i = 1, ..., n do                                 ▷ Θ(n) in total
10:         if i ≠ p then
11:             if A[i] ≤ v then
12:                 Append A[i] to A_1
13:             else
14:                 Append A[i] to A_2
15:             end if
16:         end if
17:     end for
18:     A_1' ← QuickSort(A_1)                                ▷ T(|A_1|)
19:     A_2' ← QuickSort(A_2)                                ▷ T(|A_2|)
20:     return Concatenate(A_1', [v], A_2')                  ▷ Θ(n)
21: end function
```

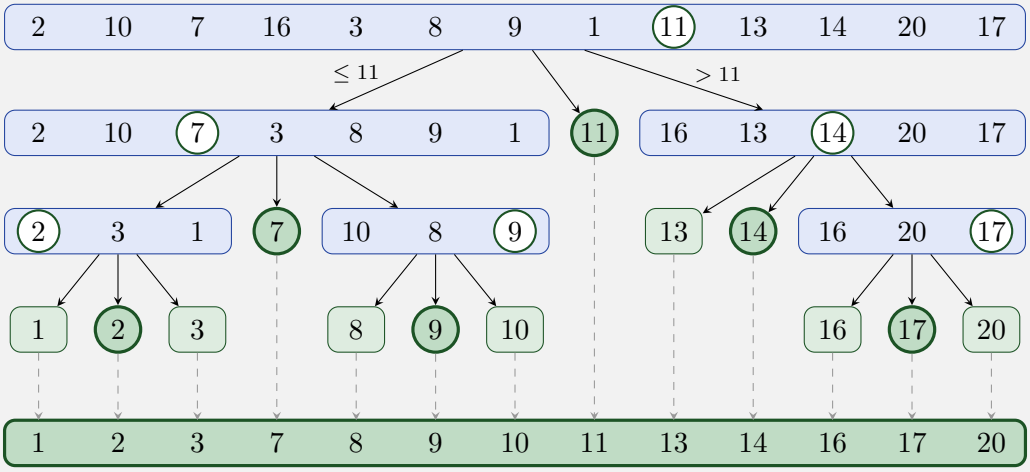
The test $i \neq p$ excludes the selected pivot from both recursive inputs. Other occurrences of the same value are kept in A_1. Each recursive input has fewer than n elements, so the algorithm eventually reaches its base cases.

11.3.2 Following the recursive calls

We now run QuickSort on the 13-element array that MergeSort sorted in Example 49.

Example 51: QuickSort on the 13-element array

Each row is one level of recursive calls, and the circled values are the random pivots. Once a pivot or a one-element array is placed, it never moves again: the dashed arrows carry it to the sorted result.



11.4 The cost of QuickSort

For the running-time analysis in this section, assume that the n values are distinct. Partitioning compares the pivot with the other $n-1$ elements. If k values are smaller than the pivot, the two recursive sizes are k and $n-1-k$. A call therefore has the form

$$T(n) = T(k) + T(n-1-k) + \Theta(n).$$

The linear term includes partitioning and the final concatenation. The random pivot determines k, so it determines how balanced the two recursive inputs will be.

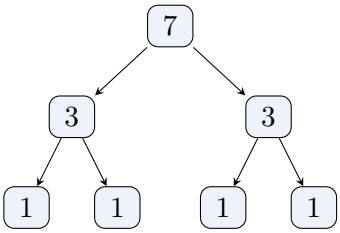
11.4.1 Balanced partitions and the worst case

Balanced partitions. A pivot near the middle of the sorted order leaves two inputs of roughly half the size. Repeated balanced partitions give logarithmic depth, with at most linear work across each level. This leads to $\Theta(n \log n)$ time.

The worst case. If the pivot is always an extreme value (the smallest or the largest), one recursive input is empty and the other has $n-1$ elements. The next nonempty call has size $n-1$, then $n-2$, and so on. The number of pivot comparisons is

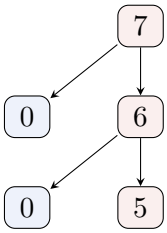
$$(n-1)+(n-2)+\cdots+1=\frac{n(n-1)}{2}=\Theta(n^2).$$

Balanced pivots



Subproblem sizes shrink quickly.

Always an extreme pivot



⋮

The nonempty branch loses only one element per call.

When there are $m \geq 2$ distinct values, the chance of picking an extreme value is $2/m$. Consequently, the probability of choosing an extreme pivot at every non-base call is

$$\frac{2}{n}\cdot\frac{2}{n-1}\cdots\frac{2}{2}=\frac{2^{n-1}}{n!}.$$

This probability tends to 0 very quickly as n grows. Randomization allows this bad execution, but does not force it.

11.4.2 The expected cost

Averaging over the random pivot choices, one can prove that the expected running time of QuickSort is $O(n \log n)$:

$$\boxed{\mathbb{E}[T(n)]=O(n\log n), \qquad T_{\text{worst}}(n)=\Theta(n^2).}$$

We do not prove the expected bound here.

12

Data structures

Start of lecture 29 October

12.1 Returning to the top-k problem

We have already met the *top-k problem* in text mining. In Section 9.9, a search engine assigns a score to each document and returns the k documents with the largest scores. Section 9.9.1 sorts all scores, while Section 9.15 uses an inverted index to reduce the number of documents scored. The last step of Algorithm 9.2 still asks us to select the K highest-scoring documents.

Now we focus on that selection step. Suppose the scores have already been computed: how can we select the best k without sorting every score?

Problem: Top-k selection

Input: n numerical scores and an integer $1 \leq k \leq n$.

Output: k entries with the largest scores, in any order. If scores are tied at the cutoff, any k entries satisfying this condition are acceptable.

Example 52: Three results out of seven

For the scores

$$12,\ 7,\ 20,\ 3,\ 16,\ 9,\ 25,$$

the top 3 are $20, 16, 25$.

First approach: sort everything. MergeSort orders the n scores in $O(n \log n)$ time; we then take the largest k. This works, but also orders scores that we will discard.

Second approach: keep only k candidates in an array. Keep the first k scores. For each new score, scan the candidate array to find its minimum. If the new score is larger, replace that minimum. Otherwise discard the new score.

Example 53: The candidate we need to replace

Suppose the current candidates are $[12, 7, 20]$. When 16 arrives, we replace 7, obtaining $[12, 16, 20]$. When 9 arrives, we keep the candidates unchanged: even the smallest candidate, 12, is better than 9.

Each scan costs $\Theta(k)$, giving $O(nk)$ time overall. For small k this can help, but finding the same kind of candidate repeatedly is expensive. We want a way to keep the smallest retained score easy to reach, while updating the candidates quickly.

This motivates *data structures*. An array is only one way to organize the candidates. Other data structures support the same operations with different costs, each offering its own advantages and limitations. We will compare these alternatives to see whether they offer a better trade-off for our problem.

12.2 An interface and its implementation

Think of a car. We can describe it through the operations it supports: starting, accelerating, braking and steering. We specify what these operations do without specifying the mechanisms that make them work.

A petrol car and an electric car support the same operations, but carry them out using different internal components. To drive either car, we need to understand the controls, not how those components work.

Definition 12.1: Abstract data type (ADT)

An *abstract data type* (i.e. the car in the example above) specifies a collection of data, the operations available on it, and their meaning. We write ADT for short.

Definition 12.2: Data structure

A *data structure* (i.e. in the example above, the specific way the car is built, using a petrol engine or an electric motor) provides a concrete implementation: it determines how the data are stored and how the operations are performed.

The same ADT can have several implementations. All must give the specified results, but their operation costs can differ. For example, a petrol car and an electric car are both cars: they support the same driving operations, even though their internal mechanisms differ.

12.3 The list ADT

Definition 12.3: List (ADT)

A *list* stores a sequence of n elements. Using positions $1, \ldots, n$, it supports:

1. get(i): return the element at position i;
2. set(x, i): replace the element at position i with x;
3. insert(x, i): insert x at position i, shifting the old positions $i, \ldots, n$ one place to the right;
4. remove(i): remove the element at position i, shifting later positions one place to the left.

For insertion, $1 \leq i \leq n + 1$; the other operations require $1 \leq i \leq n$.

The index i describes the current position in the sequence.

12.3.1 Arrays: reach a position directly

Definition 12.4: Array

An *array* is a **data structure** that stores elements in equal-sized slots occupying consecutive memory locations. It provides one possible implementation of the list ADT.

The ADT tells us which operations must be supported; the array specifies how the elements are stored so that we can implement those operations.

If the first slot starts at address b and each slot uses w bytes, slot i starts at

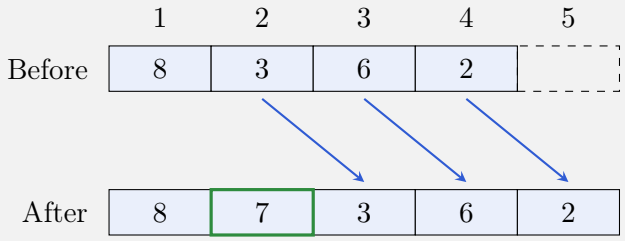
$$b + (i - 1)w.$$

The address can be calculated directly, so accessing a valid position takes $\Theta(1)$ time.

1. get(i) takes $\Theta(1)$ time: calculate the address of slot i and read its element.
2. set(x, i) takes $\Theta(1)$ time: calculate the address of slot i and replace its element with x.
3. insert(x, i) takes $\Theta(n)$ time in the worst case: inserting at the beginning shifts all n elements one place to the right. We assume that a spare slot is available.

Example 54: Inserting into an array

Insert 7 at position 2 of $[8, 3, 6, 2]$.



Move 2, then 6, then 3 to the right; write 7 in slot 2.

Moving from right to left prevents overwriting a value before it has been copied. We assume that one spare slot is available.

4. remove(i) takes $\Theta(n)$ time in the worst case: removing the first element shifts the remaining $n - 1$ elements one place to the left.

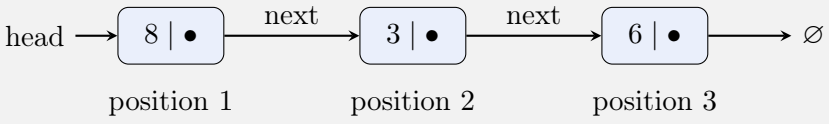
12.3.2 Linked lists: follow references

Definition 12.5: Singly linked list

A *singly linked list* is a data structure that implements the list ADT using nodes. Each node contains an element and a reference to the next node. A *head* reference identifies the first node; the last node has no successor.

Example 55: A singly linked list

The sequence $[8, 3, 6]$ is stored in three nodes.



The nodes need not occupy consecutive memory locations.

To reach position i, we start at the head and follow $i - 1$ links.

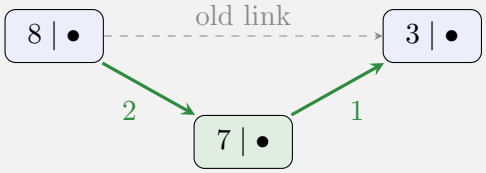
1. get(i) takes $O(n)$ time in the worst case: follow the links to position i and read the element. With a reference to the node already available, reading it takes $\Theta(1)$ time.
2. set(x, i) takes $O(n)$ time in the worst case: follow the links to position i and replace the element with x. With a reference to the node already available, the replacement takes $\Theta(1)$ time.

3. insert(x, i) takes $O(n)$ time in the worst case: first find the node preceding position i. Once that reference is available, inserting the new node takes $\Theta(1)$ time. Inserting at the head also takes $\Theta(1)$ time.

Example 56: Inserting after a known node

Suppose we already have a reference to the node containing 8. To insert 7 immediately after it, create a new node and update two references:

$$new.next \leftarrow node8.next, \qquad node8.next \leftarrow new.$$



The update takes $\Theta(1)$ time. If we were only given an index, we would first have to find the preceding node.

4. remove(i) takes $O(n)$ time in the worst case: first find the node preceding position i. With that reference available, bypass the removed node by changing the predecessor's `next` link in $\Theta(1)$ time. Removing the head takes $\Theta(1)$ time: update the head reference.

12.3.3 Doubly linked lists and operation costs

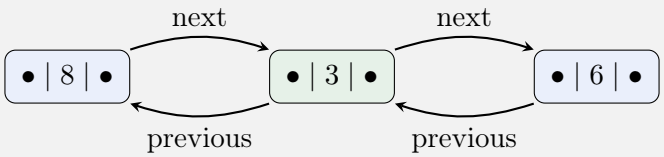
Definition 12.6: Doubly linked list

A *doubly linked list* is a data structure that implements the list ADT using nodes. Each node contains an element, a `next` reference to its successor, and a `previous` reference to its predecessor, when these exist.

From a node, we can directly reach both neighbours. Finding a node by its index still requires following links.

Example 57: A doubly linked list

The sequence $[8, 3, 6]$ has links in both directions between adjacent nodes.



From the middle node, both neighbours are directly accessible.

1. get(i) takes $O(n)$ time in the worst case to locate position i. Reading the element of a node whose reference is already available takes $\Theta(1)$ time.
2. set(x, i) takes $O(n)$ time in the worst case to locate position i. Replacing the element of a known node takes $\Theta(1)$ time.
3. insert(x, i) takes $O(n)$ time in the worst case to locate the insertion position. Given a reference to the adjacent node, linking in the new node takes $\Theta(1)$ time: update the neighbours' `next` and `previous` references, and the head or tail if necessary.
4. remove(i) takes $O(n)$ time in the worst case to locate position i. Given a reference to that node, unlinking it takes $\Theta(1)$ time: its `previous` reference gives direct access to its predecessor. Update the neighbours' links and the head or tail if necessary.

The following table concerns the *indexed operations of the list ADT*. Bounds are worst-case bounds for a list of n elements; array insertion assumes available capacity.

Data structure	get(i)	set(x, i)	insert(x, i)	remove(i)
Array	$\Theta(1)$	$\Theta(1)$	$O(n)$	$O(n)$
Singly linked	$O(n)$	$O(n)$	$O(n)$	$O(n)$
Doubly linked	$O(n)$	$O(n)$	$O(n)$	$O(n)$

The advantage of linked lists appears when the needed references are already available: inserting after a known node is constant time, and a doubly linked list also removes a known node in constant time. These are different requests from finding and removing *the element at index i*.

12.4 The priority queue ADT

For top-k selection, positions in a sequence are not our main concern. We repeatedly need the candidate with an extreme score. A priority queue provides operations tailored to that task.

Definition 12.7: Max-priority queue (ADT)

A *max-priority queue* stores entries with comparable priorities and supports:

1. insert(x): add entry x;
2. peekMax(): return an entry of maximum priority without removing it;
3. extractMax(): remove and return an entry of maximum priority.

The last two operations require a nonempty queue. Equal priorities are allowed. For a queue of numbers, each number is its own priority.

Definition 12.8: Min-priority queue (ADT)

A *min-priority queue* stores entries with comparable priorities and supports:

1. insert(x): add entry x;
2. peekMin(): return an entry of minimum priority without removing it;
3. extractMin(): remove and return an entry of minimum priority.

The last two operations require a nonempty queue. Equal priorities are allowed. For a queue of numbers, each number is its own priority.

The ADT specifies the result of each operation; it does not require a particular representation.

We now consider two data structures that can implement the max-priority queue ADT: an unsorted array and a sorted array. Both support the same three operations, but their representation changes the cost of each operation.

An unsorted array. Insertion at the end is cheap, but finding the maximum requires a scan. Once found, it can be removed by replacing it with the last entry: a priority queue does not have to preserve sequence order. The scan still makes extraction linear.

A sorted array. If the array is in increasing order, the maximum is the last entry. Reading or removing it is constant time, but inserting a new entry in sorted order may require shifting $\Theta(n)$ entries.

Can we do better than this for our use case?

We would like fast access to the maximum *and* fast updates. A binary heap provides that compromise.

12.5 Binary heaps

For the numerical heaps considered here, each node's value is its priority.

Definition 12.9: Binary max-heap

A *binary max-heap* is a data structure that implements the max-priority queue ADT. It is represented as a binary tree satisfying two conditions:

1. **Shape:** every level is full except possibly the last, which is filled from left to right. Such a tree is *complete*.
2. **Order:** each parent's value is greater than or equal to the values of its children.

Definition 12.10: Binary min-heap

A *binary min-heap* is a data structure that implements the min-priority queue ADT. It is represented as a binary tree satisfying two conditions:

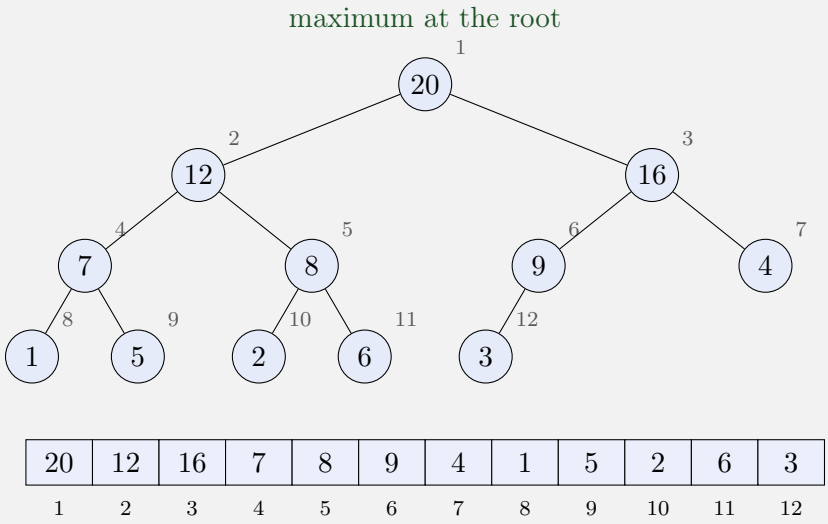
1. **Shape:** every level is full except possibly the last, which is filled from left to right. Such a tree is *complete*.
2. **Order:** each parent's value is less than or equal to the values of its children.

In a max-heap, following a path from the root downwards never increases the value. Therefore the root contains a global maximum, and peekMax() takes $\Theta(1)$ time.

In a min-heap, values never decrease along a path from the root downwards. The root therefore contains a global minimum, and peekMin() takes $\Theta(1)$ time.

Example 58: A max-heap and its array

The small labels give positions in *level order*: read each level from left to right, starting at the root.



20	12	16	7	8	9	4	1	5	2	6	3
1	2	3	4	5	6	7	8	9	10	11	12

For example, $12 \geq 7, 8$ and $16 \geq 9, 4$. The last level is filled on the left. The next inserted node occupies position 13, the right child of position 6.

Because the shape has no gaps, we can store a heap in an array $H[1], \ldots, H[n]$, as in Example 58.

12.5.1 Store the tree in an array

With *one-based indexing*, the links are implicit:

$$\text{parent}(i) = \left\lfloor \frac{i}{2} \right\rfloor \quad (i > 1), \qquad \text{left}(i) = 2i, \qquad \text{right}(i) = 2i + 1.$$

A child exists only when its index is at most n. For example, node 5 has children 10 and 11; node 6 has child 12, but no child 13 yet. We calculate these positions without storing explicit pointer fields.

At depth d, a full level has 2^d nodes. A complete tree with $n \geq 1$ nodes has height $\lfloor \log_2 n \rfloor$, counting edges from the root to the deepest leaf. A path up or down therefore contains only $O(\log n)$ edges for $n \geq 2$.

12.5.2 Insertion: move up until the order is restored

How do we insert a value in the heap while preserving the structure?

First append the new entry in the next free array slot. This preserves completeness, but its value might exceed its parent's. While that happens, swap the two and continue from the parent. This operation is called *sift-up*.

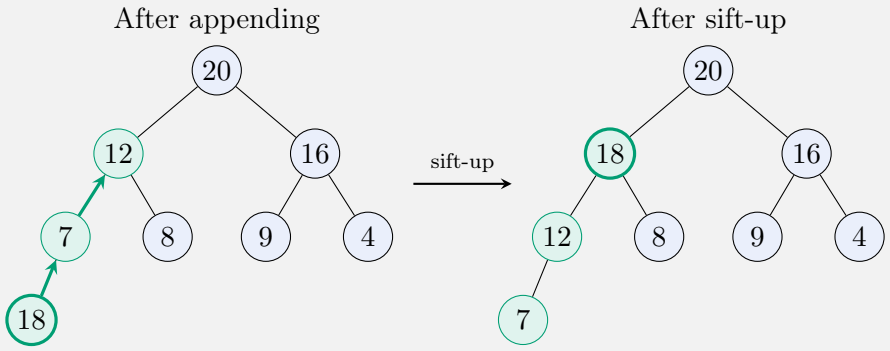
Only the new entry's path to the root can violate the order. Each swap moves it one level up, so there are at most $O(\log n)$ swaps.

Example 59: Insert 18 into a smaller max-heap

Start with $H = [20, 12, 16, 7, 8, 9, 4]$. Append 18 at position 8. Its path to the root is $8 \to 4 \to 2 \to 1$.

Action	Array
Append 18	$[20, 12, 16, 7, 8, 9, 4, \mathbf{18}]$
Swap with 7	$[20, 12, 16, \mathbf{18}, 8, 9, 4, 7]$
Swap with 12	$[20, \mathbf{18}, 16, 12, 8, 9, 4, 7]$
Stop: $18 \leq 20$	$[20, 18, 16, 12, 8, 9, 4, 7]$

Both drawings show the entire heap. Green arrows show how 18 moves up.



The root remains 20. Insertion rearranges the path without sorting the whole array.

The heap repair costs $O(\log n)$ in the worst case. With an array of sufficient capacity, this is also the insertion cost. With a dynamic array that occasionally resizes, the overall

insertion bound is $O(\log n)$ amortized for $n \geq 2$; a resizing insertion can itself take $\Theta(n)$ time.

12.5.3 Extraction: replace the root and move down

To extract the maximum:

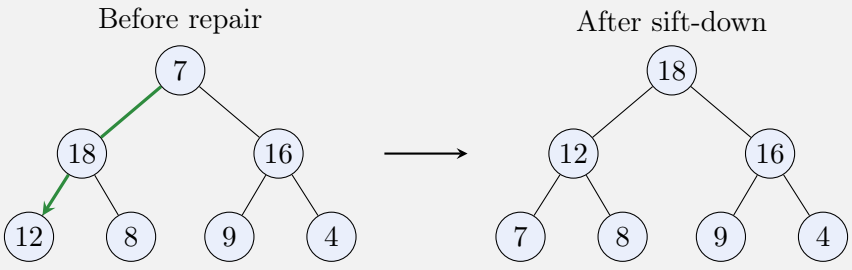
1. Save the root's entry to return it.
2. Move the last entry to the root and reduce the heap size by one. If the heap is now empty, stop.
3. If the entry is smaller than a child, swap it with its *larger* child. Continue downwards until the order is restored.

This repair is called *sift-down*. Choosing the larger child makes the new parent at least as large as both children. The possible violation moves down one path, which has logarithmic length.

Example 60: Extract 20 after the insertion

The heap from the previous example is $[20, 18, 16, 12, 8, 9, 4, 7]$. Save 20 and put the last entry, 7, at the root. Position 8 is no longer in the heap.

Action	Array
Replace root with 7	$[\mathbf{7}, 18, 16, 12, 8, 9, 4]$
Swap with 18	$[18, \mathbf{7}, 16, 12, 8, 9, 4]$
Swap with 12	$[18, 12, 16, \mathbf{7}, 8, 9, 4]$
Stop at a leaf	$[18, 12, 16, 7, 8, 9, 4]$



At the first step, choose 18 rather than 16. At the second, choose 12 rather than 8. The returned entry is 20; the new maximum is 18.

Extraction takes $O(\log n)$ time in the worst case, assuming removing the final array slot does not trigger resizing. Reading the maximum alone requires no repair and remains $\Theta(1)$.

12.5.4 Comparing priority queue implementations

For a queue containing $n \geq 2$ entries, the following bounds assume arrays have sufficient capacity and removal does not resize them. The sorted array keeps its maximum at the last occupied slot.

Representation	insert(x)	peekMax()	extractMax()
Unsorted array	$\Theta(1)$	$\Theta(n)$	$\Theta(n)$
Sorted array	$O(n)$	$\Theta(1)$	$\Theta(1)$
Binary max-heap	$O(\log n)$	$\Theta(1)$	$O(\log n)$

The heap keeps enough order to expose the maximum, without paying to keep every entry in sorted order after each insertion.

12.6 Sorting with a heap

A max-heap gives us the largest remaining value at every extraction. Can we use this operation to sort an entire array? We first use a separate output array, then see how to reuse the input array itself.

12.6.1 Sorting into a second array

Let $A[1], \ldots, A[n]$ be the input. First turn A into a max-heap, using its own storage. One way to do this with the operations we already know is to grow a heap one entry at a time: for $j = 2, \ldots, n$, regard $A[j]$ as the newly appended entry and sift it up inside $A[1], \ldots, A[j]$. The later entries are untouched. We call this procedure BuildMaxHeapByInsertion(A); it takes $O(n \log n)$ time and can be implemented with a loop using $O(1)$ extra working space.

Once the max-heap is ready, allocate a second array B of size n. For $n \geq 1$, fill it as follows:

1. For $j = n, n-1, \ldots, 2$, extract the maximum, repairing the remaining heap by sifting down, and store the extracted value in $B[j]$.
2. Copy the only remaining value into $B[1]$.

The largest value goes into the last position, the next largest into the preceding position, and so on: $B[1] \leq \cdots \leq B[n]$. There are $n - 1$ extractions; the last value needs no repair. An empty input is already sorted.

Example 61: Fill the output array from right to left

Reuse the seven-entry heap from the start of Example 59, before inserting 18:

$$H = [20, 12, 16, 7, 8, 9, 4].$$

Each row below shows B after storing one value. The symbol $\square$ denotes an output cell that has not yet been filled.

Action	Output array B
Extract 20	$[\square, \square, \square, \square, \square, \square, 20]$
Extract 16	$[\square, \square, \square, \square, \square, 16, 20]$
Extract 12	$[\square, \square, \square, \square, 12, 16, 20]$
Extract 9	$[\square, \square, \square, 9, 12, 16, 20]$
Extract 8	$[\square, \square, 8, 9, 12, 16, 20]$
Extract 7	$[\square, 7, 8, 9, 12, 16, 20]$
Copy 4	$[4, 7, 8, 9, 12, 16, 20]$

The six extractions return values in decreasing order; placing them from right to left produces an increasing array.

Building the heap takes $O(n \log n)$ time with the method above. The $n - 1$ extractions take another $O(n \log n)$ time in total, so the overall bound is $O(n \log n)$. The second array uses $\Theta(n)$ additional space.

12.6.2 Reuse the input array: in-place HeapSort

Can we avoid allocating a second array? Each extraction reduces the active heap size by one. We can store the extracted maximum in the cell that has just left the heap.

Definition 12.11: In-place sorting

An *in-place sorting algorithm* rearranges the entries in the input array using $O(1)$ additional working space. The storage occupied by the input array itself is not counted as additional space.

Suppose the active heap occupies $A[1], \ldots, A[m]$. Instead of removing its root and allocating space elsewhere:

1. Swap $A[1]$ with $A[m]$. The maximum is now in its final position, m.
2. Reduce the active heap size to $m - 1$. The physical array still contains n cells.
3. Sift down the new root, considering only positions $1, \ldots, m - 1$. The sorted suffix must stay outside the repair.

Repeat for $m = n, n - 1, \ldots, 2$. This is *HeapSort*: the heap shrinks on the left while the sorted suffix grows on the right.

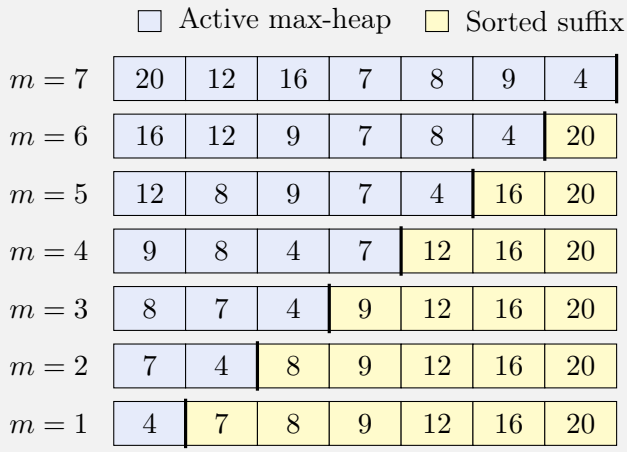
Example 62: HeapSort in the same array

Start from the same heap $[20, 12, 16, 7, 8, 9, 4]$. In the first pass, swap 20 with 4:

$$[\underbrace{4, 12, 16, 7, 8, 9}_{\text{repair this prefix}} \mid 20].$$

Sift-down swaps 4 with 16, then with 9, giving $[16, 12, 9, 7, 8, 4 \mid 20]$.

Below, every row after the first shows the *completed repair* of one pass. All seven cells belong to the same array throughout.



At $m = 1$, the remaining heap consists only of the minimum, 4. The entire array is now sorted; no further extraction is needed.

Algorithm 12.1: In-place HeapSort

```
1: function HeapSort(A)
2:     n ← length(A)
3:     if n ≤ 1 then
4:         return A
5:     end if
6:     BuildMaxHeapByInsertion(A)
7:     for m = n, n − 1, . . . , 2 do
8:         swap A[1] and A[m]
9:         SiftDownPrefix(A, 1, m − 1)
10:    end for
11:    return A
12: end function
```

Here SiftDownPrefix(A, i, m) repairs the heap starting at position i, using only the first m cells. Its children must already root valid max-heaps, as they do after the root swap in HeapSort. An iterative implementation makes the space bound explicit:

Algorithm 12.2: Sift down inside the active prefix

```
1: function SiftDownPrefix(A, i, m)
2:     while 2i ≤ m do
3:         j ← 2i                                   ▷ left child exists
4:         if j < m then                            ▷ right child also exists
5:             if A[j + 1] > A[j] then
6:                 j ← j + 1
7:             end if
8:         end if
9:         if A[i] ≥ A[j] then
10:            return
11:        end if
12:        swap A[i] and A[j]
13:        i ← j
14:    end while
15: end function
```

Why it works. At the start of a pass with active size m:

1. $A[1], \ldots, A[m]$ form a max-heap.
2. $A[m+1], \ldots, A[n]$ are in increasing order.
3. Every value in the heap is less than or equal to every value in the sorted suffix, when that suffix is nonempty.

The root is the largest remaining value. Moving it to position m extends the sorted suffix by one cell. Sift-down restores the heap in the smaller prefix without touching that suffix. When only one heap entry remains, these properties imply that the entire array is sorted. Equal values cause no difficulty.

Time and space. With the insertion-based construction described above, building the heap costs $O(n \log n)$. The sorting passes cost

$$\sum_{m=2}^{n} O(\log m) \;=\; O(n \log n).$$

Thus HeapSort takes $O(n \log n)$ time in the worst case. Using iterative sift-up and sift-down, it needs only a constant number of indices and a temporary value for swaps: $O(1)$ extra working space. The second array B is no longer needed.

12.7 Use a heap to select the top k

We can now complete the selection step introduced in Section 9.9. To keep the k *largest* scores, we repeatedly need the *smallest retained score*: it is the weakest candidate and the one a better entry replaces.

We therefore use a *min-heap* of at most k entries (Definition 12.10). Its array representation is the same as for a max-heap. Insertions sift up when a child is smaller than its parent; extraction sifts down using the smaller child. The operation bounds are unchanged.

Note: Which heap should we use?

To keep the k largest values, use a **min-heap**: its root is the smallest value currently retained, so it is the candidate to replace when a larger value arrives.

To keep the k smallest values, use a **max-heap**: its root is the largest value currently retained, so it is the candidate to replace when a smaller value arrives.

The root should expose the retained candidate that we would discard first. It is not necessarily the best result we eventually return.

Algorithm 12.3: Keep the k largest scores

```
1:  function TopKWithHeap(A, k)
2:      Q ← an empty min-priority queue with capacity k
3:      for each entry x in A do
4:          if size(Q) < k then
5:              Q.insert(x)
6:          else if x > Q.peekMin() then
7:              Q.extractMin()
8:              Q.insert(x)
9:          end if
10:     end for
11:     return the entries stored in Q
12: end function
```

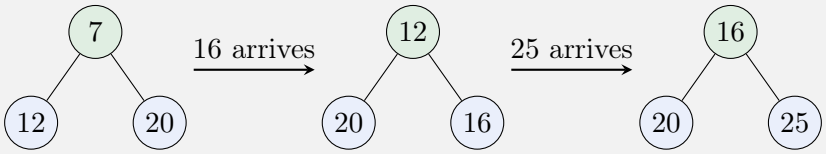
Why it works. After processing t entries, Q contains the largest $\min(k, t)$ scores seen so far, allowing any choice among tied entries. Until it fills, it keeps every entry. Afterwards, let m be its minimum. A new score $x \leq m$ cannot improve the retained scores; a score $x > m$ must replace one occurrence of m. Both cases preserve the claim. At $t = n$, it gives exactly the required result.

Example 63: Follow the top-3 candidates

Process the scores from our opening example. Arrays below are valid min-heap representations; they are not sorted lists.

New score	Action	Heap array afterwards
12	Insert	[12]
7	Insert and sift up	[7, 12]
20	Insert	[7, 12, 20]
3	Discard: $3 \leq 7$	[7, 12, 20]
16	Replace 7	[12, 20, 16]
9	Discard: $9 \leq 12$	[12, 20, 16]
25	Replace 12	[16, 20, 25]

The three full heaps at the replacement steps are:



Each root is the smallest of the three retained scores.

The final candidates have scores $16, 20, 25$. If the search engine must display them in decreasing score order, sort just these three.

12.7.1 Time, space and the search-engine connection

Each score requires a constant-time size or root check. If retained, it causes at most one extraction and one insertion on a heap of size at most k. Thus the total selection cost is

$$O\big(n \log(k+1)\big), \qquad \text{with } O(k) \text{ additional space.}$$

For $k \geq 2$, we usually write this as $O(n \log k)$. Writing $\log(k+1)$ also covers $k = 1$: keeping a single maximum requires one linear scan, not zero work. Capacity k can be allocated once; no heap growth beyond that capacity is needed. If an ordered output is required, sorting the k retained entries adds $O(k \log k)$ time.

For example, with $n = 10^6$ and $k = 10$, heap updates traverse a tree of height at most $\lfloor \log_2 10 \rfloor = 3$, rather than ordering a million scores. Every input score is still examined.

In Algorithm 9.2, the inverted index reduces the work needed to *compute* the scores. The heap improves the subsequent step that *selects* the best results. If c candidate documents receive scores, replacing their full sort with a heap changes that selection cost from $O(c \log c)$ to $O(c \log(k+1))$, for $1 \leq k \leq c$. We still need to compute each candidate's complete score before comparing it with the heap root. If fewer than k candidates are available, return all of them.

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12.8 Finding a term in the inverted index

Let us go back to the inverted index of Definition 9.10. Take three documents:

$$1: \ A\ B\ D, \qquad 2: \ B\ A\ B, \qquad 3: \ B.$$

For each term, the index stores the documents that contain it:

$$A \to [1, 2], \qquad B \to [1, 2, 3], \qquad D \to [1].$$

To answer a query we start from a term, for example B, and we must reach its list $[1, 2, 3]$. So how do we do this? What we need is a *dictionary of terms*: a structure that, given a term, returns what is stored for it.

What happens if the data are so many that the dictionary does not fit in memory? Then we would need parallel frameworks or external algorithms, which we will not discuss now.

Here we ask a more basic question: what exactly is a dictionary, and how can we build one?

12.9 The dictionary ADT

As with lists and priority queues, we first say *what* a dictionary does, and only then *how* to implement it.

Definition 12.12: Dictionary (ADT)

A *dictionary* (or *map*) D stores pairs (k, v), where the key k belongs to a set K of keys and the value v to a set V of values. Each key appears at most once. D supports:

1. insert(k, v): associate the value v with the key k;
2. get(k): return the value associated with k;
3. remove(k): remove k and its value;
4. exists(k): return true if k is in D, false otherwise.

In Python, `D["Maria"] = 7` inserts the key `"Maria"` with value 7; if `"Maria"` is already in D, its value becomes 7. To see whether `"Maria"` is in D, we can compute exists(`"Maria"`).

The most direct implementation stores the pairs in a list, $[(k_1, v_1), (k_2, v_2), \ldots]$. To see whether a key exists, or to read its value, we must scan the keys one by one. Can we reach a key directly instead?

12.10 When the keys are positions

Suppose the keys are student IDs $0, 1, \ldots, m-1$, and the value of each ID is the student's record. Then the dictionary can simply be an array (Definition 12.4) with one slot per ID:

$$D[63] = (\texttt{"Maria"}, \texttt{"De Paolis"}, \ldots).$$

The key *is* the position, so every operation reaches its slot directly, in $\Theta(1)$ time.

The problem appears when the keys are not numbers, but names or tax codes. Now the key is no longer 63, but `"Maria"`. (We assume the keys are unique, so there is only one Maria.) How can we still use an array?

12.11 Turning a key into a position

One idea is to convert the name into a number. Assign a number to each letter and add them up: this gives a function

$$f : \text{strings} \to \mathbb{N},$$

and we store each name at the position given by its number:

$$\text{insert}(name, x): \quad A[f(name)] \leftarrow x.$$

Example 64

Number each letter by its position in the alphabet: $a = 1$, $i = 9$, $m = 13$, $r = 18$. Then

$$f(\texttt{"Maria"}) = 13 + 1 + 18 + 9 + 1 = 42,$$

so Maria's record goes to $A[42]$.

There are several problems with this idea.

- **Different keys can get the same number.** Names made of the same letters have the same sum: for example, $f(\texttt{"Amira"}) = 1 + 13 + 9 + 18 + 1 = 42$, so Amira's record would also go to $A[42]$, the slot already holding Maria. Different letters can also add up to the same value: $3 + 1 = 4$ and $2 + 2 = 4$.
- **How long must the array be?** Long names give large numbers, so the array must reach the largest number a name can produce. Although this is necessary, consider the case where we store only 10 names. Almost all of the array's slots stay empty: we would allocate a huge array just for those 10.

We would like a function that sends the keys into a *small* range of positions, and that sends different keys to the same position as rarely as possible. This is what a hash function does.

12.12 Hash functions

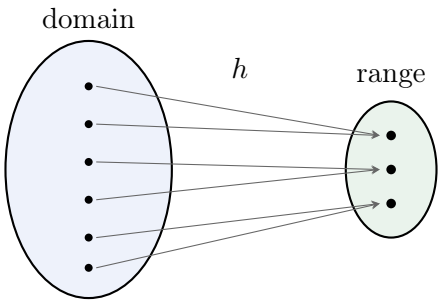
Definition 12.13: Hash function

A *hash function* is a mapping from a domain to a range,

$$h : \{0, \ldots, 2^m - 1\} \to \{0, \ldots, 2^n - 1\},$$

where typically $m \gg n$: the domain is much larger than the range.

For example, with $m = 4$ the domain contains the 16 bit strings from 0000 (that is, 0) to 1111 (that is, 15).



Another example is a hash function from $\{0, \ldots, 9999\}$ to $\{0, \ldots, 99\}$:

$$h(x) = x \bmod 100,$$

which keeps the last two digits: $3783 \to 83$. Remember that, in binary, taking a number modulo 2^k keeps its last k bits.

Since the domain is much larger than the range, many keys must share the same position.

Definition 12.14: Collision

A *collision* occurs when two different keys $x \neq y$ have the same hash value, $h(x) = h(y)$.

Collisions cannot be avoided, but a good hash function has a small number of them.

Example 65: Counting IP addresses

We want to count how many times each IP address appears, for example with `count["151.100.37.83"] += 1`. An IP address has 32 bits, and we want to send it to a range of 8 bits:

$$\{0, \ldots, 2^{32} - 1\} \to \{0, \ldots, 2^8 - 1\}.$$

If h keeps only the last byte, then 151.100.37.83 and 11.23.161.83 both go to 83: a collision. Every pair of addresses ending with the same number collides.

12.12.1 How rare should collisions be?

Write $M = 2^m$ for the size of the domain and $N = 2^n$ for the size of the range. On average, M/N keys of the domain share each position, so collisions are certain.

Example 66

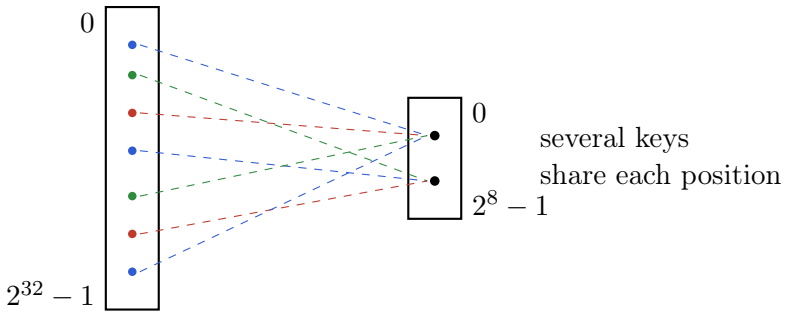
Take $m = 4$ and $n = 2$: the $M = 16$ keys $0, \ldots, 15$ go to the $N = 4$ positions $0, 1, 2, 3$. With $h(x) = x \bmod 4$:

Position	Keys
0	$0, 4, 8, 12$
1	$1, 5, 9, 13$
2	$2, 6, 10, 14$
3	$3, 7, 11, 15$

Each position receives $4 = 16/4 = M/N$ keys.

What we can ask is that no pair of keys is more unlucky than the others: for all $x \neq y$,

$$\mathbb{P}\big(h(x) = h(y)\big) = \frac{1}{2^n} = \frac{1}{N}.$$



But where does the probability come from? A probability needs something random, but here nothing is:

- the keys x and y are fixed;
- h is a fixed function.

Then $h(x) = h(y)$ is either always true or always false.

We could let h choose a random position every time we call it. Then two keys would share a position with probability $1/N$. But the dictionary would stop working. Suppose insert(`"Maria"`, x) draws position 42 and stores Maria's record in $A[42]$. Later, get(`"Maria"`) draws again, say position 17, and looks in $A[17]$: Maria is not there. To find what insert stored, get must look in the same position, so $h(x)$ must return the *same* value every time we call it on the same x. The hash function must be deterministic.

12.13 Universal hash families

The way out is to use randomness only once, at the start: we *choose at random a deterministic function* from a family of hash functions, and from then on we always use that function.

Definition 12.15: Universal hash family

A *universal hash family* is a set of hash functions from which we choose one at random, so that for every pair of keys $x \neq y$ the probability, over this choice, that $h(x) = h(y)$ is as small as the size of the range allows.

Example 67: One universal family for IP addresses

There are many ways to build a universal hash family. Here, let us try to construct one.

An IP address x is made of four bytes, $x = x_1.x_2.x_3.x_4$, and we want to send it to the range $0, \ldots, 255$.

1. Choose a prime number p larger than 255 but close to it, for example $p = 257$.
2. Choose $a_1, a_2, a_3, a_4 \in \{0, \ldots, p-1\}$ uniformly at random.
3. Set
$$h_a(x_1, x_2, x_3, x_4) = \Big(\sum_{i=1}^{4} a_i x_i\Big) \bmod p.$$

Each choice of $a_1, \ldots, a_4$ gives one function of the family. Its values lie in $\{0, \ldots, p-1\}$, that is 257 positions, just above the 256 we wanted.

Proposition 12.1: Collision probability

For every pair of addresses $x = x_1.x_2.x_3.x_4$ and $y = y_1.y_2.y_3.y_4$ with $x \neq y$,

$$\mathbb{P}\big(h_a(x) = h_a(y)\big) = \frac{1}{p},$$

where the probability is over the random choice of $a_1, \ldots, a_4$.

Proof. Since $x \neq y$, the two addresses differ in at least one byte; say $x_1 \neq y_1$ (the argument is the same for any other byte). The two addresses collide when

$$\sum_{i=1}^{4} a_i x_i \equiv \sum_{i=1}^{4} a_i y_i \pmod{p}, \quad \text{that is} \quad a_1(x_1 - y_1) \equiv -\sum_{i=2}^{4} a_i (x_i - y_i) \pmod{p}.$$

Imagine that a_2, a_3, a_4 have already been drawn. Then the right-hand side is a fixed number, and only a_1 is still random. Write

$$d = x_1 - y_1, \qquad c = -\sum_{i=2}^{4} a_i(x_i - y_i),$$

so that the equation becomes $a_1 d \equiv c \pmod{p}$, with $a_1 \in \{0, \dots, p-1\}$. How many values of a_1 satisfy it?

Step 1: p does not divide d. Each byte lies in $\{0, \dots, 255\}$, so $-255 \le d \le 255$, and $d \neq 0$ because $x_1 \neq y_1$. The only multiple of $p = 257$ in this range is 0, so $p \nmid d$.

Step 2: different values of a_1 give different remainders. Take $a, b \in \{0, \dots, p-1\}$ with $ad \equiv bd \pmod{p}$. Then $(a-b)\,d \equiv 0 \pmod{p}$, that is, p divides the product $(a-b)\,d$. A prime that divides a product divides one of its factors, and by Step 1 it does not divide d, so $p \mid (a-b)$. But $-(p-1) \le a - b \le p-1$, and the only multiple of p in this range is 0. Hence $a = b$.

Step 3: exactly one value of a_1 works. By Step 2, the p values $a_1 = 0, \dots, p-1$ give p different remainders $a_1 d \bmod p$. There are only p possible remainders, $0, \dots, p-1$, so each of them is reached exactly once. In particular, exactly one value of a_1 gives the remainder of c.

Step 4: the probability. a_1 is drawn uniformly from p values, independently of a_2, a_3, a_4, and only one of them makes the two addresses collide. So, once a_2, a_3, a_4 are fixed, the collision has probability $1/p$. This holds whatever values a_2, a_3, a_4 took: c changes, but by Step 3 there is always exactly one right value of a_1. Averaging over all choices of a_2, a_3, a_4 therefore gives

$$\mathbb{P}\big(h_a(x) = h_a(y)\big) = \frac{1}{p}. \qquad \square$$

12.14 Hash tables

We can now implement the dictionary. Keep an array A of p slots and store the value of key z in slot $A[h(z)]$. For example:

$$z_1 \to 135, \quad z_2 \to 60, \quad z_3 \to 23, \quad z_4 \to 80, \quad z_5 \to 60, \quad z_6 \to 75, \quad z_7 \to 111.$$

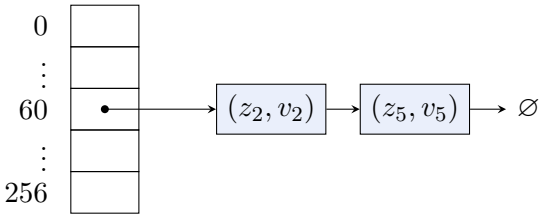
Here z_2 and z_5 collide. Collisions are rare, but they can happen.

If there are no collisions, finding an element takes constant time:

$$\text{get}(z): \quad \text{return } A[h(z)].$$

But with a collision, two values would need to be stored in the same slot. How can we keep both?

The simplest way is to store a linked list (Definition 12.5) in each slot: slot 60 holds the pairs of all the keys that hash to 60.



Now get(z) goes to slot $A[h(z)]$ and scans its list until it finds the key z. The cost depends on the length of that list, so the time stays constant when the lists are short.